

Atomic beam simulations and polarization determination using the Breit-Rabi polarimeter

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emphasize that this is about re-evaluation of the existing polarimeter systematics and the way
forward to achieve $\Delta Q/Q \leq 1\% \rightarrow \Delta P/P \leq 1\% \Rightarrow$ maybe adopt the title accordingly

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I. QUESTIONS / TO-DOS

Below a list of questions and items we should address **Please provide text and figures as needed. Many of the answers shall become part of the writeup.**

A. Atomic beam simulations

1. Distributions and optics

- Velocity distributions at: target, BRP (Breit-Rabi polarimeter), and transition units.
- Radial/azimuthal profiles at target and BRP:

- i. Are the distributions uniform?
- ii. Do they show structure vs. azimuthal angle, are they axially symmetric around the beam axis?
- iii. How do profiles change if, e.g., the last ABS magnet is transversely displaced?
- (c) Transverse acceptance:
 - i. What is the accepted transverse-velocity distribution?
 - ii. Would it make sense to add a transverse velocity to the longitudinal one?

2. Physics models and parameters

- (a) Attenuation parameter:
 - i. What is it?
 - ii. How does changing it affect the results?
 - iii. What attenuation options exist?
- (b) Density and flux:
 - i. Which equations are used to calculate the density?
 - ii. How is the conversion from density to flux done in the code?
- (c) Magnetics:
 - i. How is the pole-tip field calculated from the magnetization J of the materials?
 - ii. What are the numerical values of the transmissions for states 1-4 to the target and to the BRP detector using the 1.5 T pole tip field and the updated pole tip field values?
 - iii. Need a table with parameters for each magnet, inner radius at entrance, inner radius at exit, outer radius, length.
- (d) Beam kinematics:
 - i. Why is the maximum velocity generated not exactly 1807 m/s as in the input file?

3. Event generator

- (a) How exactly is the event generator working?
- (b) Does each simulation start from the same seed to eliminate statistical fluctuations?
- (c) What are its different options doing?
- (d) Is the event generator described in the HJET design paper?
- (e) Ideally, as there are no transition units inside the ABS, the output parameters of the atomic beam exiting the last magnet of the ABS should be identical, i.e., have identical transmission for each of the four states. Is this the case?

II. OVERVIEW

ABSSim simulates a neutral atomic beam traversing skimmers and sextupole magnets. It generates many independent particle *tracks* characterized by position $= (x, y, z)$ and velocity $v = (v_x, v_y, v_z)$ together with an internal hyperfine (spin) state s . Each track is sampled from configurable source models (“Starting Generators”) and then propagated through a longitudinal mesh using either straight drift or Runge–Kutta integration in magnetic-field regions. Along the path, the code records acceptance, density maps, losses, velocity spectra, and optional RF-induced state flips.

III. KEY INPUTS & SYMBOLS

- Gas type: hydrogen or deuterium (`run.gastype`).
- Internal states: initial hyperfine states (s).
- Source generator choice (`inten.SGflag`): 1=Extended, 2=Molecular, 3=Effusive, 4=DSMC file, 5=Other file.

- Source geometry: nozzle $z = \text{layout.zs}$, skimmer radii/positions, magnet apertures.
- Velocity distribution parameters (`vdist`): drift/peak $b_0 = \text{vdist.vd}$, “type” exponent `ftype`, beam temperature $T_b = \text{vdist.btem}$.
- Random seed (`StGen.seed`); number of tracks `StGen.Ntracks`.

IV. RANDOM NUMBERS & WEIGHTS

- RNG is seeded in `RanGenInit()` via `srand()`, using `StGen.seed` (if > 0) or wall-clock time.
- Each generated track carries weight $w_2 = 1/\text{Ntracks}$; per-state weight is $w = w_2/\text{run.nstates}$. For density observables proportional to flux, a factor $w_3 = w/v_z$ is used where appropriate.

V. ANGLE GENERATION

Angles are expressed by $\cos \theta = v_z/|v|$ and azimuth $\phi = \text{atan2}(v_y, v_x)$.

- **Polar angle θ :** `GetPolarAngle()` samples $x = \cos \theta$ with PDF $\propto x^n dx$ on $[\text{inten.cthlim}, 1]$, where $n = \text{vdist.icos}$ and $\text{inten.cthlim} = \cos(\theta_{\max})$ encodes the largest geometrically allowed angle set by the current generator and aperture.
- **Azimuth ϕ :** depends on the Starting Generator:
 - *Extended (1)*: uniform $\phi \sim \mathcal{U}(-\pi, \pi)$.
 - *Effusive (3)*: $v_y = 0$ (motion in x - z plane).
 - *Molecular (2)*: $\phi = \text{atan2}(y_2 - y_0, x_2 - x_0)$ set by straight line from a sampled nozzle point to a sampled skimmer point.
 - *Other (5)*: read (θ, ϕ) from file.
 - *DSMC (4)*: read (v_x, v_y, v_z) from file (no separate ϕ sampling).

VI. VELOCITY DISTRIBUTION AND SAMPLING

The velocity distribution of the generated atoms is defined through the input file `inputfiles/current.veldis`, which provides both the parameters for the speed distribution $f(v)$ and for the initial angular spread of the beam. A typical example is:

```
name: inputfiles/current.veldis
ICOS
  1.0
VDRIFT
  1807.0
BEAMTEMP
  20.0
CC      FUNC
2
```

Each keyword corresponds to an internal variable in `vdist` or `vconst`, as described below.

a. ICOS This value sets the exponent in the angular weighting function used when sampling the initial polar angle θ . In the code, it is stored as `vdist.icos` and used in `GetPolarAngle()` to generate

$$P(\cos \theta) \propto (\cos \theta)^{\text{vdist.icos}}, \quad \text{inten.cthlim} \leq \cos \theta \leq 1.$$

A larger ICOS value makes the distribution more strongly peaked along the beam axis, while ICOS=0 corresponds to a uniform angular spread within the allowed cone.

b. VDRIIFT The drift velocity parameter (`vdist.vd`) sets the peak or characteristic velocity b_0 in the longitudinal direction. It controls the most probable speed of the atoms but not necessarily the mean value of the distribution. When the beam temperature is nonzero (see below), the actual average velocity $\langle v \rangle$ will exceed VDRIIFT due to thermal broadening.

c. BEAMTEMP The translational temperature of the beam, in kelvins. It is stored as `vdist.btem` and determines the spread of the velocity distribution. The corresponding Gaussian width parameter

$$a_0 = \frac{m}{2k_B T_{\text{beam}}}$$

enters directly into the analytic form of the velocity probability density function.

d. FUNC The keyword `FUNC` specifies the functional type of the distribution and is stored as `vdist.ftype`. In most cases (`FUNC=2`), the one-dimensional speed probability density used by `GetVel()` and `vtable()` is

$$f(v) \propto v^{(\text{ftype}+1)} \exp[-a_0(v-b_0)^2],$$

where a_0 and b_0 are computed in `vinit()`. The exponent `ftype + 1` adjusts the low-velocity tail: higher `FUNC` increases the weight of larger speeds.

e. Resulting Mean Velocity For these parameters, the model computes the normalized distribution and its mean in `vinit()` as

$$\langle v \rangle = \frac{b_0}{2} \left(1 + \sqrt{1 + \frac{2(\text{ftype} + 1)}{a_0 b_0^2}} \right),$$

so that even with `VDRIIFT=1807` m/s and `BEAMTEMP=20` K, the calculated average startup velocity is typically $\langle v \rangle \approx 2.0 \times 10^3$ m/s.

f. Sampling Procedure The steps performed by the code are:

- (a) The constants a_0 , b_0 , and normalization N_v are computed in `vinit()`, and a lookup table for the inverse cumulative distribution is built in `vtable()`.
- (b) During track generation, `GetVel()` draws a random number $u \in (0, 1)$, finds the corresponding bin in the inverse-CDF table, and linearly interpolates to obtain a continuous speed v .
- (c) The sampled v is then combined with a random direction (θ, ϕ) from `GetPolarAngle()` and the generator-specific azimuth rule (see Section ??) to form

$$v_z = v \cos \theta, \quad v_x = v \sin \theta \cos \phi, \quad v_y = v \sin \theta \sin \phi.$$

g. Physical Interpretation The combination of these parameters allows the user to reproduce a wide range of atomic-beam conditions: a cold beam with narrow thermal spread (small `BEAMTEMP`), a collimated beam (large `ICOS`), or a hotter, more divergent source (small `ICOS`, large `BEAMTEMP`). The mean speed and angular distributions defined here set the initial conditions for all subsequent track propagation in `CalcTrack()`.

VII. POSITION GENERATION (BY STARTING GENERATOR)

Extended (1):: Sample (x_0, y_0) uniformly in a disk of radius `ExtStGen.Rnoz` at $z_0 = \text{layout.zs}$; sample (θ, ϕ) as above.

Effusive (3):: Point-like virtual nozzle at $z_0 = \text{layout.zs}$; ϕ fixed ($v_y = 0$).

Molecular (2):: Sample a point in nozzle disk and a point in upstream skimmer disk; ray from the first to the second sets direction (θ, ϕ) . Start at z_0 by linear interpolation of (x_0, y_0) along that ray.

DSMC (4):: Read x and (v_x, v_y, v_z) at plane $z = \text{FileStGen.Zpos}$, then translate back to z_0 by drift.

Other (5):: Read (x, y, θ, ϕ) per track from file.

VIII. TRACK PROPAGATION

Propagation is handled in `CalcTrack()` across a uniform z -mesh (bin size `pmesh.dpl`):

- (a) At each step, compute $\Delta t = \text{pmesh.dpl}/v_z$.
- (b) Determine if the step lies in field-free region (`fmap.it`) or inside a magnet:
 - *Drift*: update (r, v) with straight-line kinematics (`strght()`).
 - *Magnet*: integrate Lorentz-like spin-dependent forces via Runge-Kutta (`rukuc()`), using the local field map (`fmap`, magnet geometry).
- (c) Apply radial aperture checks; if outside, mark a loss (`lhitmap()`), terminate track.
- (d) Record:
 - Acceptance fraction (`pmesh.accept`) with weight w .
 - Density hits into (z, r) bins (`denshitmap()`, optional counter `dens_count`).
 - Optional monitors at special z -planes (`mhitmap()`).
- (e) **RF transitions (optional)**: for each enabled RF unit, detect crossing of its mid-plane; with efficiency P_{RF} , flip s between configured pair (hydrogen: `hrft[0] ↔ hrft[1]`, deuterium: `drft[0] ↔ drft[1]`). Each RF triggers at most once per track.
- (f) Compression-tube acceptance: if $r \leq r_{\text{tube}}$, accumulate longitudinal acceptance profile `acptrz.acceptrz`.

Propagation stops when the end of mesh is reached or the particle is lost.

IX. DENSITY AND FLUX IN ABSSIM

What is computed

ABSSim determines both the spatial **number density** $n_{i,j}$ (atoms/m³) and the **axial flux** $\Phi_{i,j}$ (atoms/s or atoms/s/m²) along the beamline. These quantities are related by the steady-state continuity equation:

$$\Phi = \iint n v_z dA, \quad \Phi_{i,j} = n_{i,j} \bar{v}_{z,i,j} A_j,$$

where $A_j = 2\pi r_j \Delta r$ is the annular ring area and $\Delta V_{i,j} = A_j \Delta z$ is the corresponding volume cell.

Numerical formulation

Each track carries a weight w proportional to the atomic rate it represents (atoms/s). When a particle travels an axial step Δz with velocity v_z , it spends a residence time $\Delta t = \Delta z/v_z$ in the current cell. The Monte-Carlo estimator for the expected number of atoms in that cell is

$$S_{i,j} = \sum_{\text{steps in } (i,j)} w \frac{\Delta z}{v_z}.$$

The number density and corresponding flux are then

$$\boxed{n_{i,j} = \frac{S_{i,j}}{\Delta V_{i,j}} = \frac{1}{A_j} \sum_{\text{steps in } (i,j)} \frac{w}{v_z}, \quad \Phi_{i,j} = n_{i,j} \bar{v}_{z,i,j} A_j.}$$

How the code implements it

a. Per-step weighting. In the generator modules (`IntenExtended`, `IntenMolecular`, `IntenEffusive`, etc.), the weights are defined as

$$\text{trk.wt2} = \frac{1}{\text{Ntracks}}, \quad \text{trk.wt} = \frac{\text{trk.wt2}}{\text{run.nstates}}, \quad \boxed{\text{trk.wt3} = \frac{\text{trk.wt}}{v_z}}.$$

Here `trk.wt` (atoms/s) is the per-state rate weight, and `trk.wt3` (atoms-s/m) is the residence-time weight used for volumetric density tallies.

b. State multiplicity. Each trajectory is repeated once for every active hyperfine state:

- For a single-state run (1, 2, 3, or 4), `run.nstates` = 1 and `trk.wt` = `trk.wt2`.
- For a multi-state configuration (1+2+3+4), `run.nstates` = 4 and `trk.wt` = `trk.wt2`/4.

c. Depositing into (z, r) cells. Each propagation step in `CalcTrack()` calls

```
denshitmap(rpos, lt-1, s2.is);
```

where the radial bin index is $i_r = \lfloor r/\text{dens.dplr} \rfloor$, and the raw accumulator is updated as

```
dens.sumall[lt-1][i_r] += trk.wt3.
```

This stores the total residence-time-weighted rate w/v_z for each (z_i, r_j) bin. A parallel array `dens.sumacpt` accumulates contributions only from trajectories that reach specific acceptance boundaries.

d. Normalization to physical units. After all tracks have been propagated, the routine `densnorm()` scales the dimensionless sums into physical number densities using the expression:

```
norm = 2.0 * constant.pi * dens.dplr * dens.dplr;
qin  = inten.flin * inten.diss * 5.0E19 * (inten.facpt / norm);
norm_cal = qin / (j + 0.5);
dens.sumall[i][j] *= norm_cal;
```

The normalization factor can be written analytically as

$$n_{i,j} = \frac{Q_{\text{in}}}{2\pi r_j \Delta r} \text{dens.sumall}[i][j], \quad Q_{\text{in}} = 5 \times 10^{19} * \text{inten.flin} * \text{inten.diss} * (\text{inten.facpt}/\text{norm}).$$

Here:

- `inten.flin` — the input gas flux at the source (in mbar-l/s). In our setup, `inten.flin` = 1.25.
- `inten.diss` — the dissociation efficiency of the dissociator. In our setup, `inten.diss` = 0.85.
- `inten.facpt` — the geometric acceptance of the source, calculated from the angular cutoff and emission distribution as:

$$\text{inten.facpt} = 1.0 - (\text{inten.cthlim})^{(\text{vdist.icos}+1.0)}.$$

This factor corresponds to the integral $\int_{\cos \theta_{\text{lim}}}^1 \cos^n \theta d(\cos \theta)$, representing the accepted solid angle fraction.

- The constant 5.0×10^{19} converts mbar-l/s to atoms/s.

Direct flux tallies at monitor planes

Planar flux densities are computed independently in `mplane.c` via:

```

step = mon[ind-1].xmon / mon[ind-1].nxbin;
j     = (int)(r / step);
mon[ind-1].count[j] += trk.wt / (2.0 * constant.pi * r * step);

```

This implements the differential flux per unit area:

$$J_z(r) = \frac{d\Phi}{dA} \approx \frac{\text{trk.wt}}{2\pi r \Delta r},$$

with J_z in atoms/(s·m²). Integration over all bins yields the total flux through the plane:

$$\Phi_{\text{plane}} = 2\pi \sum_j r_j J_z(r_j) \Delta r = \sum_j \text{trk.wt}.$$

The routine `zeromp()` resets all monitor-plane accumulators.

Output and units summary

By default, the output routine `dumpdensity()` writes

```
dens.sumall[i][j] → results/density.dat
```

as the **flux** $\Phi_{i,j}$ (atoms/s). If `densnorm()` has already been called, the same array holds the number density $n_{i,j}$ (atoms/m³), related by $\Phi_{i,j} = n_{i,j} v_{z,i,j} A_j$.

- $\Delta z = \text{pmesh.dpl}$ (m), $\Delta r = \text{dens.dplr}$ (m), $A_j = 2\pi r_j \Delta r$ (m²).
- $w = \text{trk.wt}$ (atoms/s per track), $w/v_z = \text{trk.wt3}$ (atoms·s/m per track).
- n in atoms/m³ (`densnorm()`), Φ in atoms/s (`dumpdensity()`), J_z in atoms/(s·m²) (`mphitmap()`).

This ensures that volumetric densities, annular fluxes, and monitor-plane fluxes are all normalized consistently to the physical beam rate.

X. BEAM ATTENUATION MODEL

Overview

As the neutral atomic beam travels along the beamline, its intensity can decrease due to interactions with residual gas, scattering on surfaces, or beam-beam collisions. In **ABSSim**, such losses are represented by *attenuation factors*, which scale the local density and flux after all geometric acceptance effects have been applied.

The attenuation model is evaluated near the end of the `acpt()` routine through two functions:

- `avedcalc()` — computes the average attenuation per segment of the vacuum system, based on local gas pressure and temperature data.
- `CalcAtten()` — applies global multiplicative factors for two physical mechanisms: residual gas attenuation (RGA) and ion-beam scattering (IBS).

Attenuation Parameters

The model introduces two user-defined scalar parameters, stored internally as:

```

atten.attenrga: residual gas attenuation factor,
atten.attenibs: ion-beam scattering attenuation factor.

```

Each factor multiplies the transmitted flux to account for losses accumulated between the source and the measurement point. The overall transmission applied to the simulated beam is therefore

$$I(z) = I_0 \times \text{atten.attenrga} \times \text{atten.attenibs}, \quad (1)$$

where I_0 is the geometrically accepted intensity determined from the track statistics.

Interpretation of Values

atten.attenrga: models losses due to collisions of the atomic beam with the background gas molecules in the beamline vacuum. It can be expressed approximately as $\exp(-L/\lambda_{\text{mfp}})$, where L is the total path length and λ_{mfp} the mean free path. Lower vacuum pressure (or shorter beam path) gives a value closer to 1.

atten.attenibs: accounts for losses from beam-beam or beam-ion interactions (for example, collisions with the circulating proton or electron beam in the storage ring). It is typically determined empirically or from separate simulation studies.

Default / Current Configuration

In the current configuration of the ABSsim setup, both attenuation parameters are set to unity:

$$\text{atten.attenrga} = 1.0, \quad \text{atten.attenibs} = 1.0.$$

This choice corresponds to the idealized case of a perfectly transparent beamline with no scattering or absorption. Hence, attenuation does not modify either the flux density (**dens.aveATT**) or the log-averaged attenuation array (**dens.logATT**) computed in **CalcAtten()**.

Numerical Implementation

When attenuation is enabled (**atten.RGAflag** or **atten.IBSflag** set true), **CalcAtten()** performs the following steps:

- (a) Opens the specified input file (**atten.filename**) containing the chamber pressure and temperature.
- (b) Computes local attenuation ratios and populates **dens.aveATT[i]** and **dens.logATT[i]** arrays along the longitudinal mesh.
- (c) Multiplies the integrated acceptance profiles **acptrz.acceptrz[i]** by the total attenuation factor.

When both factors are 1.0, these routines are still called for bookkeeping, but they leave the results unchanged. This ensures that when realistic pressure or scattering data become available, attenuation effects can be introduced seamlessly without modifying the rest of the simulation chain.

XI. TYPICAL WORKFLOW

- (a) Choose generator mode (1–5), set source geometry and T_b , b_0 , **ftype**, and seeding.
- (b) Call **vinit()** and **vtable()** to prepare the speed distribution / inverse-CDF.
- (c) Loop over **Ntracks**: sample (x_0, y_0, z_0) , (θ, ϕ) , v ; build v ; loop over $s \in \text{run.isi}$; propagate via **CalcTrack()**.
- (d) Post-process acceptance, densities, losses; compute intensity at points of interest.

XII. NOTES

- The speed mean $\langle v \rangle$ generally exceeds the drift parameter b_0 when $T_b > 0$; set $T_b \rightarrow 0$ if you want $\langle v \rangle \approx b_0$.
- For quantities proportional to particle *current* through a plane, use $w_3 = w/v_z$.
- RF transitions are triggered at the RF *mid-plane*, once per RF per track.

XIII. VELOCITY DIAGNOSTICS FROM ABSIM

Figure 1 shows the simulated speed distributions at several z -planes along the beamline. The left panel uses linear y -scale (overall shape and peak positions), while the right panel uses logarithmic y -scale to reveal low-probability tails and weaker components more clearly.

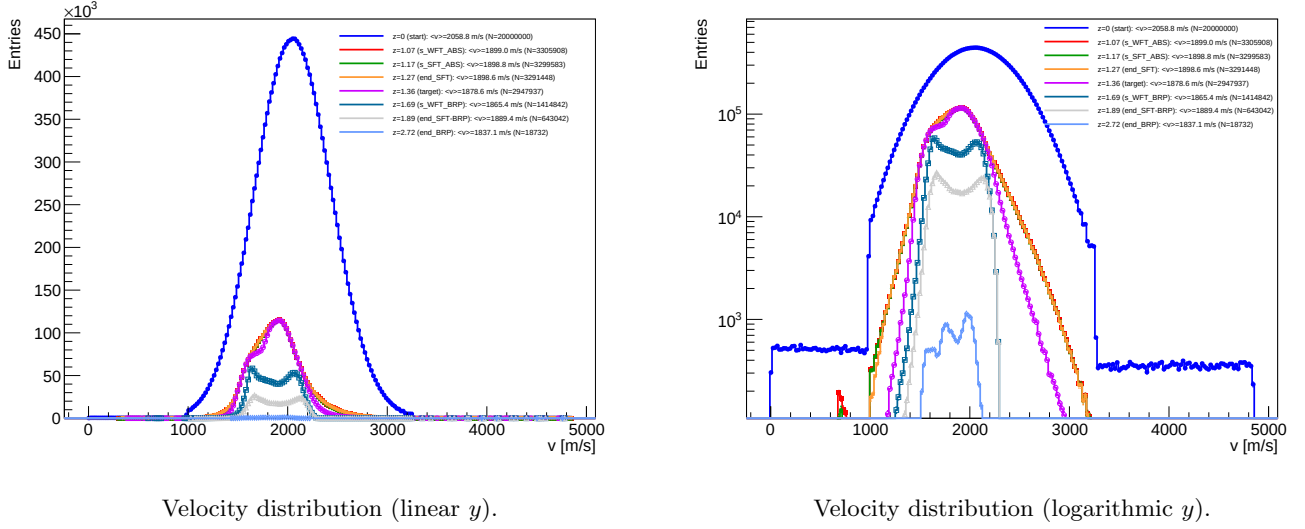


FIG. 1: Simulated velocity spectra at selected z -locations. Colors/linestyles are shared across the two panels.

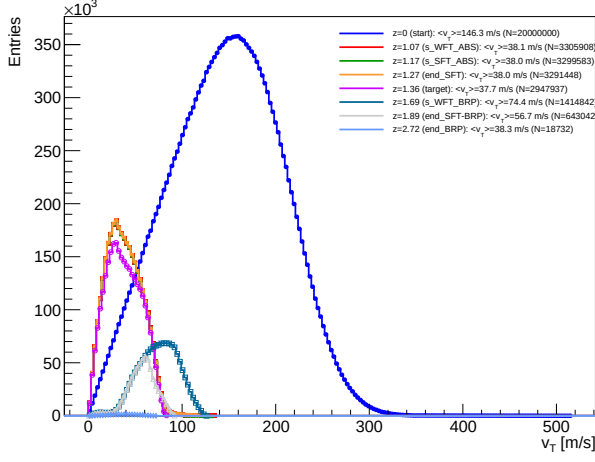
a. Planes and legend tags. Each curve corresponds to one longitudinal sampling plane; the legend entries follow the form “ $z = \text{value (label)}$ ”. For convenience, we summarize the tags used in the figures:

- **start, $z=0$ (start):** source plane at the nozzle reference ($z = 0$).
- **$z=1.07$ (s_WFT_ABS):** start of WFT (ABS section).
- **$z=1.17$ (s_SFT_ABS):** start of SFT (ABS section).
- **$z=1.27$ (end_SFT):** end of SFT.
- **$z=1.36$ (target):** target plane.
- **$z=1.69$ (s_WFT_BRP):** start of WFT (BRP section).
- **$z=1.89$ (end_SFT-BRP):** end of SFT in BRP.
- **$z=2.72$ (end_BRP):** end of BRP section.

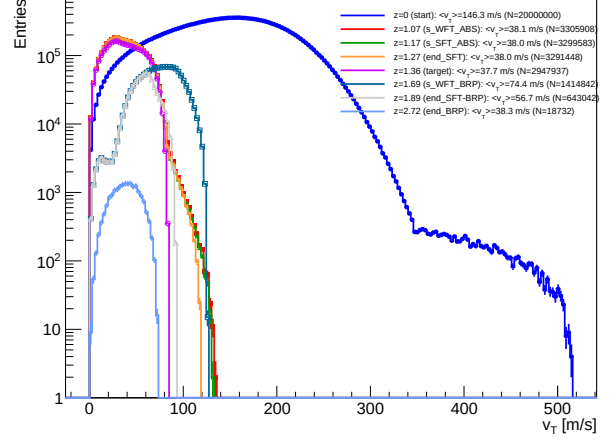
Here, the prefix **s_** denotes the *start* of a hardware element, and **end_** denotes its *end*. Comparing the linear vs. log- y panels helps identify subtle changes in the high-velocity tail and any weakly transmitted components that may be invisible on a linear scale.

A. Transverse Velocity Diagnostics

Figure 2 shows the *transverse* speed distributions $v_T = \sqrt{v_x^2 + v_y^2}$ at the same z -planes used for the total-velocity plots in Fig. 1.



Transverse velocity distribution (linear y).



Transverse velocity distribution (logarithmic y).

FIG. 2: Simulated transverse speed v_T spectra at the same z -planes as Fig. 1: start, $z=0$ (start), $z=1.07$ (s.WFT_ABS), $z=1.17$ (s.SFT_ABS), $z=1.27$ (end.SFT), $z=1.36$ (target), $z=1.69$ (s.WFT_BRP), $z=1.89$ (end.SFT-BRP), $z=2.72$ (end.BRP).

B. Polar and Azimuthal Angle Diagnostics

The angular kinematics are quantified by the polar angle $\theta = \arccos(v_z/|v|)$ and the azimuthal angle $\phi = \text{atan2}(v_y, v_x)$ (both in radians). The following figures show their distributions at the same z -planes used in Figs. 1 and 2.

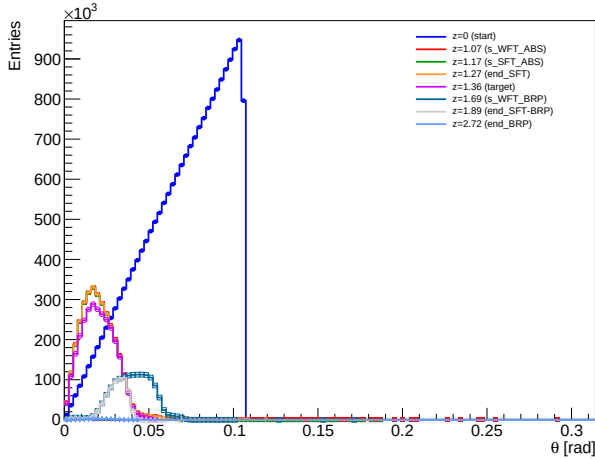
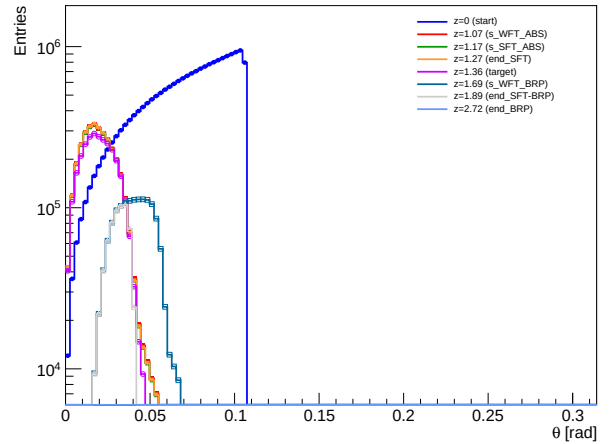
Polar angle θ distribution (linear y).Polar angle θ distribution (logarithmic y).

FIG. 3: Simulated θ spectra at the same z -planes as Fig. 1: start, $z=0$ (start), $z=1.07$ (s.WFT_ABS), $z=1.17$ (s.SFT_ABS), $z=1.27$ (end.SFT), $z=1.36$ (target), $z=1.69$ (s.WFT_BRP), $z=1.89$ (end.SFT-BRP), $z=2.72$ (end.BRP). The log- y panel highlights low-probability large- θ tails shaped by apertures and magnetic transport.

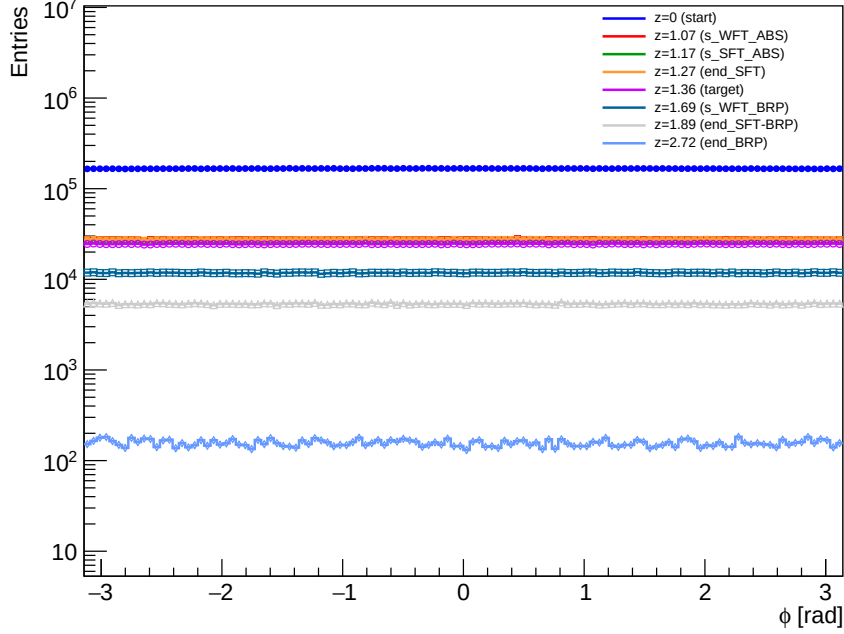


FIG. 4: Azimuthal angle ϕ distribution (radians) at the same z -planes as Fig. 1. For generator modes with axial symmetry, ϕ is uniform at the source and keep uniform along the beamline.

C. Transverse Phase-Space Footprint (X-Y)

Figure 5 shows the transverse spatial distributions x - y at the same eight z -planes used for the velocity and angle diagnostics.

4. Geometry and operational assumptions

- (a) The RF transitions should consistently take place at the longitudinal center of each transition unit.
- (b) The inner diameter of a transition unit is assumed to be $\varnothing = 30$ mm. Atoms that intercept this cylinder shall be removed from the atomic beam.
- (c) We don't have a good technical description of all aperture restrictions along the beam axis of the source. Therefore, we should extract and use the aperture restrictions as shown in Fig. 6.

5. Analysis code

- (a) Where the current code? What exactly is plotted on Fig. 8?
- (b) We have to modify the code that it measures polarization state 0 (with both WFT and SFT off)

6. Simulation variations to diagnose discrepancies

- (a) Reduce acceptance in the first magnet by a factor of 2.
- (b) Use ideal transition efficiencies of 1.
- (c) Reduce the size of beam blockers in the BRP.

BRP analysis

1. Check individual states from atomic simulations to identify the issue why with simulated beam from atomic tracking tracking, the equation system doesn't solve, while it solves correctly with known inputs.
2. Signal at target for all 16 combinations of RF transitions should be checked. Should be the same for all of them.

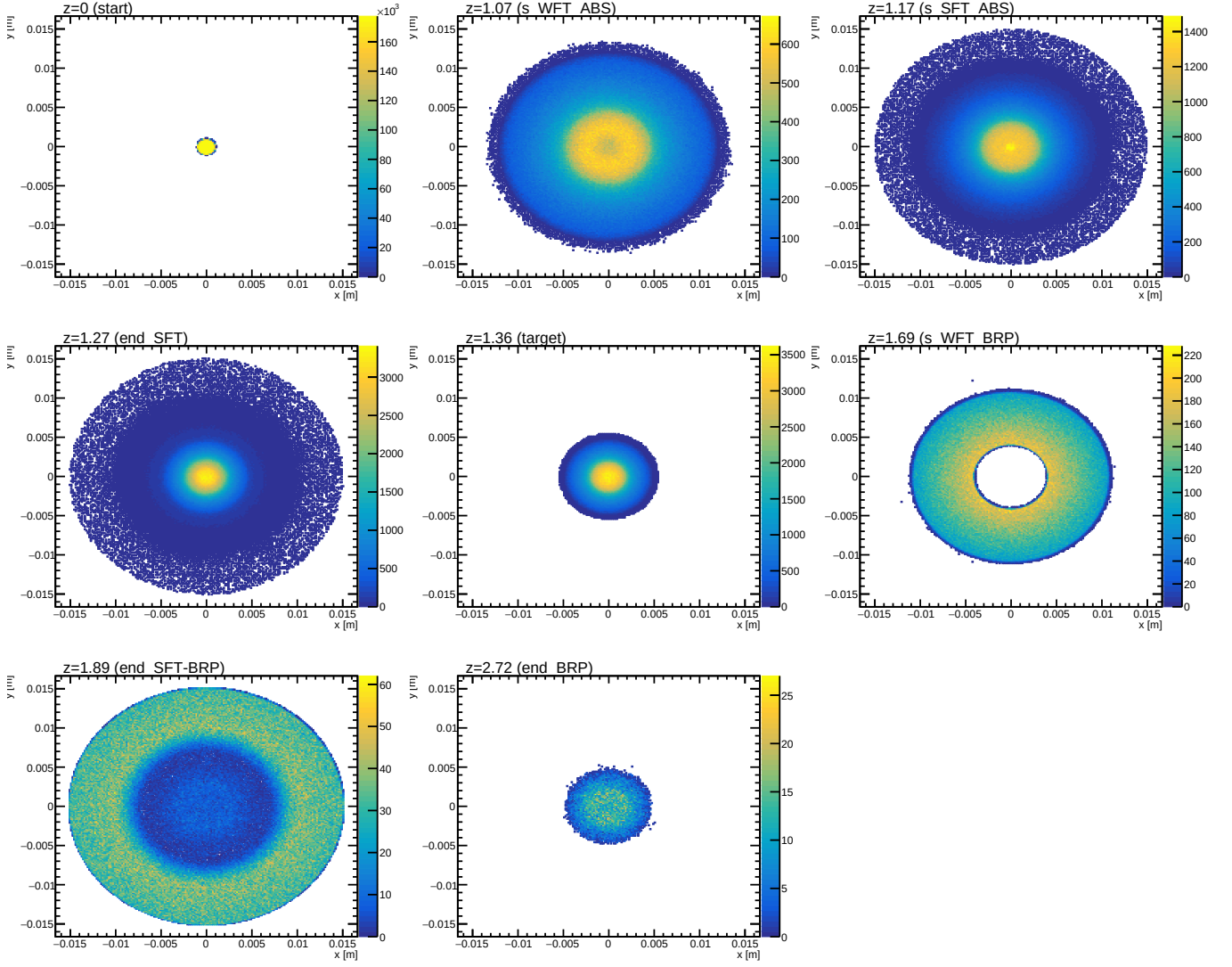


FIG. 5: Transverse x - y density (arb. units) from source to end of BRP, shown as a 3×3 mosaic with the same eight z -planes used in Figs. 1–4. Evolution of the footprint reflects angular acceptance, focusing/defocusing in sextupoles, aperture clipping and blockers at each stage.

XIV. THE OPTICAL SYSTEM OF THE ATOMIC BEAM SOURCE

A. All elements

Table I shows the relevant objects along the ABS axis.

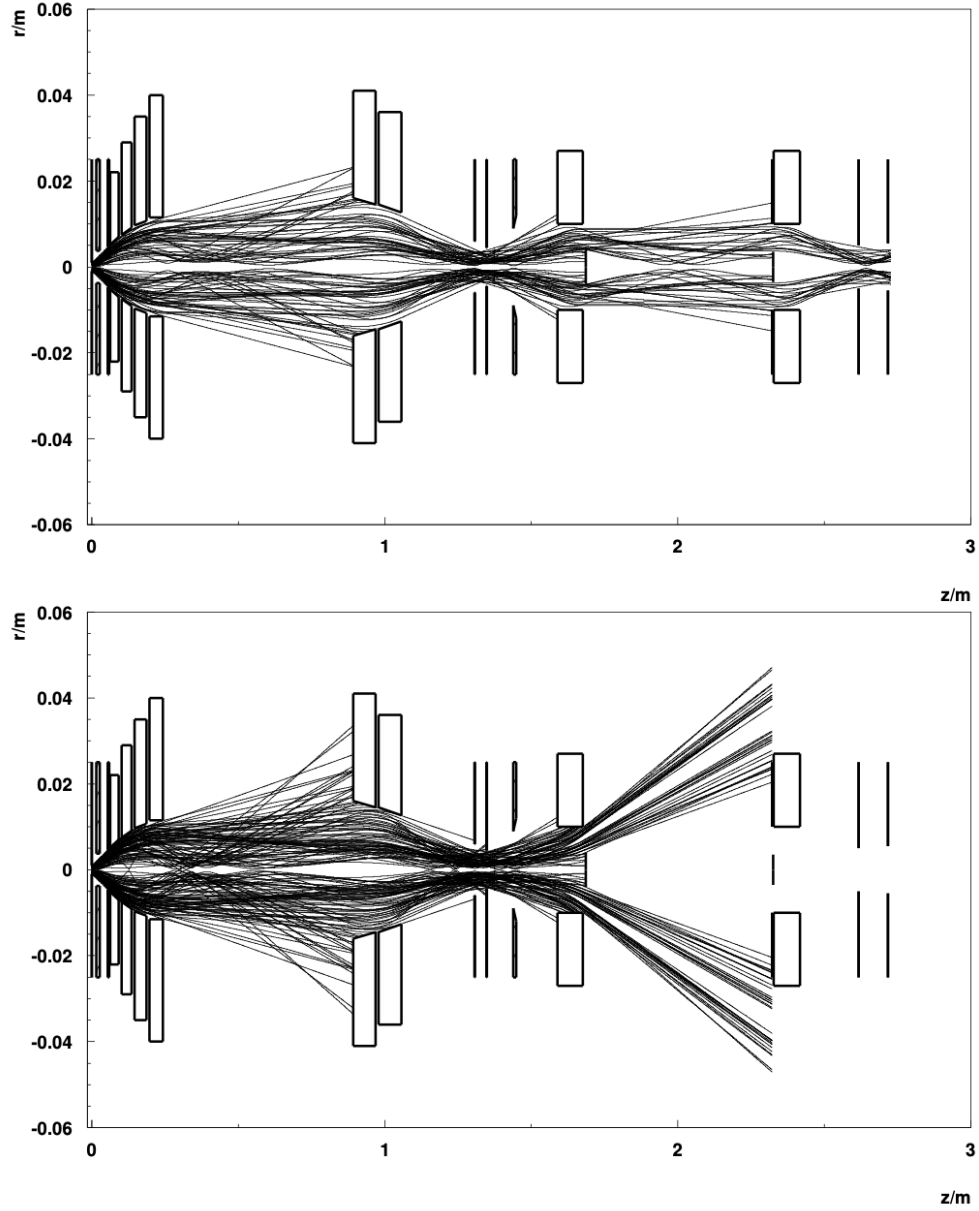


FIG. 6: Tracking calculations for the initial state $|1\rangle$ with all RF transitions off (top), and with the ABS WFT $1 \leftrightarrow 3$ transition on (bottom). (Figure taken from an Ref. [1].)

B. Magnet system of the atomic beam source

1. Calculation of pole tip fields B_0

The magnitude of the magnetic flux density inside a sextupole magnet of inner radius r_1 and pole-tip field B_0 at a distance r from the axis is given by

$$B(r) = B_0 \left(\frac{r}{r_1} \right)^2. \quad (2)$$

TABLE I: Setup information of the HJET for the tracking calculations. r_1 denotes inner radii, r_2 outer radii. Aperture 5 is the entrance of the ion gauge volume where the BRP signals are measured. Effective pole tip fields of the 8 magnets are listed in Table XIV B 1.

z [mm]	Object	Length [mm]	r_1^{entr} [mm]	r_1^{exit} [mm]	r_2 [mm]
-27.5	Nozzle	27.5	6.4	1.0	16.4
0.0	Nozzle exit	0	1.0	1.0	3.3
?18.5?	Skimmer	13.7	2.7	10.4	—
?61.2?	Collimator	0	?	?	?
55.0	Magnet 1	28.5	5.2	7.0	22.0
93.5	Magnet 2	35.4	7.5	9.1	29.0
138.9	Magnet 3	40.5	9.7	10.7	35.7
189.4	Magnet 4	46.0	11.5	11.5	40.3
885.4	Magnet 5	75.0	16.0	14.5	41.2
970.4	Magnet 6	79.0	14.5	12.7	36.7
1069.4	WFT-ABS	100.0	15.0	15.0	—
1169.4	SFT-ABS	100.0	15.0	15.0	—
1306.2	Aperture 1	0	6.0	6.0	—
1346.9	Aperture 2	0	4.5	4.5	—
1435.8	Aperture 3	0	9.0	9.0	—
1580.4	Magnet 7	87.0	10.0	10.0	27.0
1683.6	Blocker 1	0	4.0	4.0	—
1687.4	WFT-BRP	100.0	15.0	15.0	—
1787.4	SFT-BRP	100.0	15.0	15.0	—
2322.7	Blocker 2	0	3.5	3.5	—
2317.4	Magnet 8	90.0	10.0	10.0	27.0
2615.0	Aperture 4	0	5.1	5.1	—
2714.1	Aperture 5	0	5.5	5.5	—

TABLE II: Magnet system of the HJET. Columns: (1) magnet number; (2) magnetic length; (3) entrance inner radius r_1^{entr} ; (4) exit inner radius r_1^{exit} ; (5) outer radius r_2 ; (6) conical-average pole-tip field B_0^{eff} computed as in Sec. XIV B 1; (7) measured per-magnet reduction factor^a f_r and (8) corresponding reduced field $f_r B_0^{\text{eff}}$ are given here for completeness. The tracking calculations will use of B_0^{eff} .

Magnet	Length [mm]	r_1^{entr} [mm]	r_1^{exit} [mm]	r_2 [mm]	B_0^{eff} [T]	reduction f_r	$f_r \cdot B_0^{\text{eff}}$ [T]
1	28.5	5.2	7.0	22.0	1.695	0.900	1.525
2	35.4	7.5	9.1	29.0	1.686	0.930	1.568
3	40.5	9.7	10.7	35.7	1.687	0.910	1.535
4	46.0	11.5	11.5	40.3	1.687	0.920	1.552
5	75.0	16.0	14.5	41.2	1.585	0.895	1.419
6	79.0	14.5	12.7	36.7	1.584	0.875	1.286
7	87.0	10.0	10.0	27.0	1.585	0.940	1.490
8	90.0	10.0	10.0	27.0	1.585	0.925	1.466

^aThe field reduction factor has been reported by Peter Wanderer (BNL) in 2003 and was determined using an uncalibrated Hall probe.

For a cylindrical $2N$ -multipole magnet, assembled from M segments of permanently magnetized material, the pole-tip field B_0 is related to the Halbach formalism [2] by

$$B_0 = JH_3K_3, \quad (3)$$

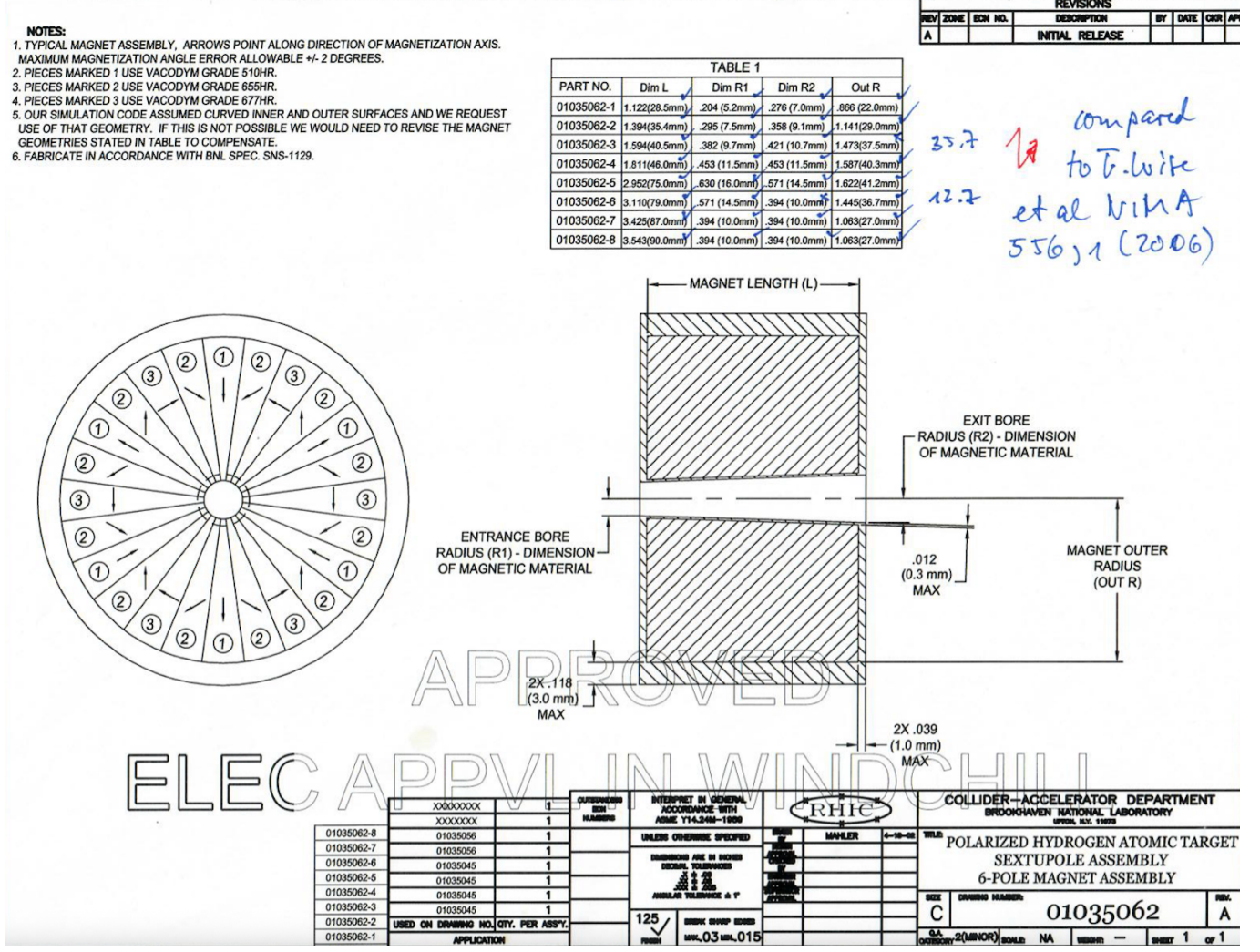


FIG. 7: Magnet assembly (left) and the dimensions of the magnetic and outer materials (right). The magnet apertures are defined by the outer cover.

where J is the remanent magnetization of the material, $n = N + \nu M = 3$ (where $\nu = 0$). The quantity H_n is defined as

$$H_n = \frac{n}{n-1} \left[1 - \left(\frac{r_1}{r_2} \right)^{n-1} \right], \quad (4)$$

where r_1 and r_2 correspond to the inner and outer radii of the magnet. The factor K_n is given by

$$K_n = \frac{\sin[(n+1)\epsilon\pi/M]}{(n+1)\pi/M}. \quad (5)$$

As discussed in Ref. [3], the sextupole magnets used in the HJET are made from three different material with varying remanence and coercivity. As indicated in [3, Fig. 1], the remanent magnetizations are

- 6/24 slices are made from VACODYM 510HR with $J = 1.41$ T,
- 12/24 slices from VACODYM 383HR with $J = 1.28$ T, and
- 6/24 slices from VACODYM 400HR with $J = 1.16$ T.

For conical bores with a linear taper over length L , we use the length average of H_3 :

$$r_1(z) = r_{1,\text{in}} + \frac{z}{L}(r_{1,\text{out}} - r_{1,\text{in}}). \quad (6)$$

$$\overline{H_3} = \frac{1}{L} \int_0^L H_3(r_1(z), r_2) dz. \quad (7)$$

For $n = 3$ this evaluates to

$$\overline{H_3} = \frac{3}{2} \left[1 - \frac{r_{1,\text{in}}^2 + r_{1,\text{in}} r_{1,\text{out}} + r_{1,\text{out}}^2}{3 r_2^2} \right]. \quad (8)$$

Hence the effective pole-tip field is

$$B_0^{\text{eff}} = J K_3 \overline{H_3}. \quad (9)$$

The filling factor ϵ , which accounts for slits between the segments required for gluing, is close to 1. Here, $M = 24$, and Table II lists the pole-tip fields B_0^{eff} of the 8 magnets of the system.

It should be noted that in the `ABSsim` simulation code, an additional parameter called the *octopole fraction* (octf) is introduced. This parameter adds an octupole field component, so the expression for $B(r)$ becomes

$$B(r) = (1 - \text{octf}) B_0 \left(\frac{r}{r_1} \right)^2 + \text{octf} B_0 \left(\frac{r}{r_1} \right)^4. \quad (10)$$

- mention somewhere that in the absence of a measurement of B_0 , what we do here is the next best thing to do.
- in practice, the magnetic fields extend before and behind the entrance and exit faces of the magnets, as shown in [3, Fig. 12]. The $\int B d\ell$ appears to be the same, but the focusing isn't quite the same. Effectively, the magnets are longer than assumed in the tracking calculations.
- what to do about the octupole components?

XV. NUCLEAR TARGET POLARIZATION AT THE INTERACTION POINT

The ultimate goal of the atomic beam system is to produce a hydrogen jet with controlled nuclear polarization at the interaction region. The nuclear polarization depends on the hyperfine state composition of the beam and the magnetic holding field B_0 at the target.

A. Nuclear polarization and population evolution through the magnet system

In the magnetic holding field B_0 at the IP, each hyperfine state has a specific nuclear polarization determined by the Breit-Rabi formula

$$Q_1(B_0) = \text{nuclear polarization of state } |1\rangle, \quad (11)$$

$$Q_2(B_0) = \text{nuclear polarization of state } |2\rangle, \quad (12)$$

$$Q_3(B_0) = \text{nuclear polarization of state } |3\rangle, \quad (13)$$

$$Q_4(B_0) = \text{nuclear polarization of state } |4\rangle. \quad (14)$$

The overall nuclear polarization of the hydrogen jet is given by the population-weighted average

$$Q = \frac{\sum_{i=1}^4 n_i \cdot Q_i(B_0)}{\sum_{i=1}^4 n_i} \quad (15)$$

where the n_i represent the population of state $|i\rangle$ at the interaction point.

The atomic beam system begins with equal populations of atoms exiting the nozzle with n_i for $i = 1$ to 4. These populations first pass through the sextupole focusing system where each state is transmitted with efficiency σ_i^A . When beam blockers in the ABS are present, these eliminate states 3 and 4, setting $\sigma_3^A = \sigma_4^A = 0$. Finally, selective RF transitions in the ABS operate with efficiencies ε_{ij}^A to create the desired hyperfine state combinations, where the superscript A identifies the transition units in the ABS.

We can represent the state populations as a column vector $\mathbf{n}$ with four entries for the hyperfine states. The resulting nuclear polarization becomes

$$Q = \frac{\mathbf{Q} \cdot \mathbf{n}}{|\mathbf{n}|} \quad (16)$$

where the row vector $\mathbf{Q} = (Q_1, Q_2, Q_3, Q_4)$ are the individual state polarizations at field B_0 . Behind the sextupole magnets and beam blockers, the state populations become

$$\mathbf{n} = \begin{pmatrix} n \\ n \\ n \\ n \end{pmatrix} \xrightarrow{\text{transitions in ABS off}} \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ 0 \\ 0 \end{pmatrix} = \mathbf{n}_{c1}^A. \quad (17)$$

XVI. SWITCHING SCHEME AND BRP SIGNALS

The state population numbers start from an equally populated system of hyperfine states at the entrance of the 1st sextupole magnet. Therefore, the vector describing the state populations is

$$\begin{pmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \end{pmatrix} = \begin{pmatrix} n \\ n \\ n \\ n \end{pmatrix}. \quad (18)$$

TABLE III: The BRP signals I_k using different combinations of transition units in the ABS. The transmissions σ_1 and σ_2 are defined for the full sextupole system from the entrance of 1st magnet to the sensor in the BRP.

Symbol	ABS-WFT	ABS-SFT	BRP signal
I_0	off	off	$(\sigma_1 + \sigma_2)n$
I_4	off	on	$(\sigma_1 + \varepsilon_{24}^A \sigma_2)n$
I_8	on	off	$(\varepsilon_{13}^A \sigma_1 + \sigma_2)n$
I_{12}	on	on	$(\varepsilon_{13}^A \sigma_1 + \varepsilon_{24}^A \sigma_2)n$

The determination of the ratio $\frac{\sigma_1 n}{\sigma_2 n}$ requires a measurement with ABS SFT, ABS WFT, and BRP WFT, measuring

$$I_{14} = (\varepsilon_{13}^A \varepsilon_{13}^B \sigma_1 + \varepsilon_{24}^A \sigma_2)n, \quad (19)$$

where ε_{13}^B is the transition efficiency of the BRP WFT. Under the assumption that the product of transition efficiencies can be neglected, from this one can obtain

$$\frac{\sigma_1}{\sigma_2} \approx \frac{I_4 - I_{14}}{I_8 - I_{12} + I_{14}} \quad (20)$$

Although four equations in Table III are written down, only three are independent. In fact, the fourth signal satisfies the exact relation

$$I_{12} = I_4 + I_8 - I_0, \quad (21)$$

which follows directly by substituting the definitions of I_0 , I_4 , and I_8 . Thus, the equation for I_{12} does not provide new information; it is a linear combination of the other three. The system therefore has rank three, leaving one degree

of freedom undetermined. To obtain a unique solution for $\sigma_1, \sigma_2, \varepsilon_{13}^A, \varepsilon_{24}^A$, additional information is required, such as an independent constraint on the ratio σ_2/σ_1 .

A. Solution 1

Therefore, an estimate of the transition efficiencies can be accomplished via evaluation of

$$\begin{aligned}\varepsilon_{13}^A &= \frac{\frac{I_8}{I_4 - I_{12}} - \frac{\sigma_2}{\sigma_1}}{1 + \frac{I_8}{I_4 - I_{12}}}, \\ \varepsilon_{24}^A &= \frac{\frac{I_4}{I_8 - I_{12}} - \frac{\sigma_1}{\sigma_2}}{1 + \frac{I_4}{I_8 - I_{12}}}.\end{aligned}\tag{22}$$

B. Solution 2

Starting from

$$r = \frac{\sigma_2}{\sigma_1}, \quad I_0 = \sigma_1 + \sigma_2,\tag{23}$$

the transmissions are

$$\sigma_1 = \frac{I_0}{1 + r}, \quad \sigma_2 = \frac{r I_0}{1 + r}.\tag{24}$$

The transition efficiencies then follow as

$$\varepsilon_{24}^A = 1 + \frac{I_4 - I_0}{\sigma_2} = 1 + \frac{(I_4 - I_0)(1 + r)}{r I_0},\tag{25}$$

$$\varepsilon_{13}^A = 1 + \frac{I_8 - I_0}{\sigma_1} = 1 + \frac{(I_8 - I_0)(1 + r)}{I_0}.\tag{26}$$

C. Which solution is more stable/better?

Solution 2 is preferred as it avoids small denominators like

$$I_4 - I_{12} \quad \text{and} \quad I_8 - I_{12},\tag{27}$$

so it is more robust numerically when experimental noise is present.

D. Target polarization

1. Not quite correct version

The target polarization for the injection of states $|1\rangle + |4\rangle$ and $|2\rangle + |3\rangle$ can be determined via

$$\begin{aligned}Q_{|1\rangle+|4\rangle}^+ &= \frac{\sigma_1 + \sigma_2 \cos(2\theta) (2\varepsilon_{24}^A - 1)}{\sigma_1 + \sigma_2}, \\ Q_{|2\rangle+|3\rangle}^- &= \frac{\sigma_1 (1 - 2\varepsilon_{13}^A) - \sigma_2 \cos(2\theta)}{\sigma_1 + \sigma_2},\end{aligned}\tag{28}$$

but the transmission should be taken from 1st magnet to target, not all the way through to the BRP detector.

2. More correct version

Therefore, a more correct expression for the target (tgt) polarization for the injection of states $|1\rangle + |4\rangle$ and $|2\rangle + |3\rangle$ is given by

$$\begin{aligned} Q_{|1\rangle+|4\rangle}^+ &= \frac{\sigma_1^{\text{tgt}} + \sigma_2^{\text{tgt}} \cos(2\theta) (2\varepsilon_{24}^A - 1)}{\sigma_1^{\text{tgt}} + \sigma_2^{\text{tgt}}}, \\ Q_{|2\rangle+|3\rangle}^- &= \frac{\sigma_1^{\text{tgt}} (1 - 2\varepsilon_{13}^A) - \sigma_2^{\text{tgt}} \cos(2\theta)}{\sigma_1^{\text{tgt}} + \sigma_2^{\text{tgt}}}. \end{aligned} \quad (29)$$

3. Fully correct version

In the absence of blockers in the ABS, the transmissions σ_3^{tgt} and σ_4^{tgt} contribute and the correct expression for the target (tgt) polarization for the injection of states $|1\rangle + |4\rangle$ and $|2\rangle + |3\rangle$ is given by

$$\begin{aligned} Q_{|1\rangle+|4\rangle}^+ &= \frac{(\sigma_1^{\text{tgt}} - \sigma_3^{\text{tgt}}) + (\sigma_2^{\text{tgt}} - \sigma_4^{\text{tgt}}) \cos(2\theta) (2\varepsilon_{24}^A - 1)}{\sigma_1^{\text{tgt}} + \sigma_2^{\text{tgt}} + \sigma_3^{\text{tgt}} + \sigma_4^{\text{tgt}}}, \\ Q_{|2\rangle+|3\rangle}^- &= \frac{(\sigma_1^{\text{tgt}} - \sigma_3^{\text{tgt}}) (1 - 2\varepsilon_{13}^A) - (\sigma_2^{\text{tgt}} - \sigma_4^{\text{tgt}}) \cos(2\theta)}{\sigma_1^{\text{tgt}} + \sigma_2^{\text{tgt}} + \sigma_3^{\text{tgt}} + \sigma_4^{\text{tgt}}}, \end{aligned} \quad (30)$$

and with $\sigma_3^{\text{tgt}} = \sigma_4^{\text{tgt}} = 0$, Eq.(29) is recovered.

E. Asymmetry in Eqs. (28)

With equal weights: $\sigma_1 = \sigma_2$, $\cos(2\theta) = 0.92120$, for $B_0 = 120$ mT, $B_c = 50.684$ mT, one obtains

$$Q^+ = \frac{1}{2}[1 + 0.92120(2\varepsilon - 1)], \quad Q^- = \frac{1}{2}[(1 - 2\varepsilon) - 0.92120]. \quad (31)$$

$$\varepsilon_{24}^A = \varepsilon_{13}^A = 1 : \quad Q^+ = 0.96060, \quad Q^- = -0.96060, \quad (32)$$

$$\varepsilon_{24}^A = \varepsilon_{13}^A = 0.99 : \quad Q^+ = 0.95139, \quad Q^- = -0.95060 \quad (33)$$

$$\varepsilon_{24}^A = \varepsilon_{13}^A = 0.98 : \quad Q^+ = 0.94218, \quad Q^- = -0.94060. \quad (34)$$

Imperfect flips ($\varepsilon < 1$) and finite mixing ($\cos(2\theta) < 1$) break antisymmetry, $Q^+ \neq -Q^-$. In fact,

$$Q^+ + Q^- = (1 - \varepsilon)(1 - \cos(2\theta)). \quad (35)$$

Therefore any symmetrized estimate is biased,

$$\bar{Q} = \frac{1}{2}(Q^+ + Q^-) = \frac{1}{2}(1 - \varepsilon)(1 - \cos(2\theta)) \neq 0, \quad (36)$$

vanishing only in the ideal limits $\varepsilon = 1$. The above equation shows that the asymmetry becomes smaller with increasing B_0 . For $B_0 = 400$ mT with equal weights $\sigma_1 = \sigma_2$, one gets

$$x = \frac{B_0}{B_c} = \frac{400}{50.684} = 7.89204, \quad \cos(2\theta) = \frac{x}{\sqrt{1 + x^2}} = 0.99207. \quad (37)$$

$$Q^+ = \frac{1}{2}[1 + 0.99207(2\varepsilon - 1)], \quad Q^- = \frac{1}{2}[(1 - 2\varepsilon) - 0.99207]. \quad (38)$$

$$\varepsilon_{24}^A = \varepsilon_{13}^A = 1 : \quad Q^+ = 0.99603, \quad Q^- = -0.99603, \quad (39)$$

$$\varepsilon_{24}^A = \varepsilon_{13}^A = 0.99 : \quad Q^+ = 0.98611, \quad Q^- = -0.98603, \quad (40)$$

$$\varepsilon_{24}^A = \varepsilon_{13}^A = 0.98 : \quad Q^+ = 0.97619, \quad Q^- = -0.97603. \quad (41)$$

F. Systematic difference in target polarizations $Q_{|1\rangle+|4\rangle}^+$ and $Q_{|2\rangle+|3\rangle}^-$ using Eqs. (29) and Eqs. (30)

Using $x = 2.367$ (Hydrogen, $B = 120$ mT), and the fully correct expressions in Eqs. (30) and the one ignoring the transmission of states $|3\rangle$ and $|4\rangle$ [Eq. (29)], we evaluate the target polarization at

$$\tan(2\theta) = \frac{1}{x}, \quad x \equiv \frac{B}{B_c} = 2.367 \quad \Rightarrow \quad \cos(2\theta) = \frac{x}{\sqrt{1+x^2}} \approx 0.921166, \quad (42)$$

with preliminary values for the transmissions from Vera/Zhengqiao

$$\sigma_1^{\text{tgt}} = 0.2935, d\sigma_2^{\text{tgt}} = 0.2899, \quad \sigma_3^{\text{tgt}} = 0.00007, \quad \sigma_4^{\text{tgt}} = 0.00027, \quad \varepsilon_{24}^A = 0.97, \quad \varepsilon_{13}^A = 0.98. \quad (43)$$

The resulting polarizations are

$$\begin{aligned} Q_{|1\rangle+|4\rangle}^+ \big|_{\text{with } 3,4} &\approx 0.9333, \quad Q_{|1\rangle+|4\rangle}^+ \big|_{\text{without } 3,4} \approx 0.9323, \quad \Delta Q^+ \equiv Q_{\text{with } 3,4}^+ - Q_{\text{without } 3,4}^+ \approx -0.0010, \\ Q_{|2\rangle+|3\rangle}^- \big|_{\text{with } 3,4} &\approx -0.9307, \quad Q_{|2\rangle+|3\rangle}^- \big|_{\text{without } 3,4} \approx -0.9296, \quad \Delta Q^- \equiv Q_{\text{with } 3,4}^- - Q_{\text{without } 3,4}^- \approx +0.0011. \end{aligned} \quad (44)$$

Thus, when the contributions of states $|3\rangle$ and $|4\rangle$ are included in the target transmission, the polarization values are shifted relative to the simplified two-state treatment by $\Delta Q^{+,-}$. This corresponds to a systematic difference of about one percentage point in magnitude for both Q^+ and Q^- at $B = 120$ mT. At $x = 2.367$ ($B = 120$ mT), the true target polarizations for the “+” and “−” settings differ by ≈ 0.008 , and each is lower by $\approx 1.2\%$ relative to the case that neglects transmission of states $|3\rangle$ and $|4\rangle$ (i.e., $\sigma_3^{\text{tgt}} = \sigma_4^{\text{tgt}} = 0$), corresponding to a systematic relative deviation of about 1.3% for both Q^+ and Q^- .

In this regime, *averaging the target polarizations* is not indicated: the finite transmission of $|3\rangle$ and $|4\rangle$ shifts Q^+ and Q^- by roughly $\pm 1.3\%$ in opposite directions, so the arithmetic mean $\frac{1}{2}(Q^+ + Q^-)$ appears artificially stable and *masks the true, configuration-dependent systematics*. One should report Q^+ and Q^- separately rather than the average.

XVII. POLARIZATION ANALYSIS CODE HJET

Figure 8 shows what the current analysis code puts out. The crucial info about intensity when ABS WFT and ABS SFT are off, is currently not recorded, as if transition efficiencies would be of no interest.

XVIII. TRANSMISSION THROUGH THE HJET BEAMLINE WITH BEAM BLOCKERS INSTALLED IN BRP

Let us denote the transmission through the ABS sextupole system as A, through the first BRP magnet as B1, and through the second as B2. Two beam blockers are installed in the BRP: one immediately after the first magnet and the second in front of the second magnet. Here we assume that the blockers completely suppress the signals from states 3 and 4.

Let us write down the transmissions through the entire HJET beamline for 4 cases, listed in Table III and additionally a case with both ABS transitions on and a BRP-WFT on.

A. Transmission to the target

There are four possible modes for atoms traveling from the nozzle to the target: both ABS transitions off, only ABS-WFT on, only ABS-SFT on, and both transitions on.

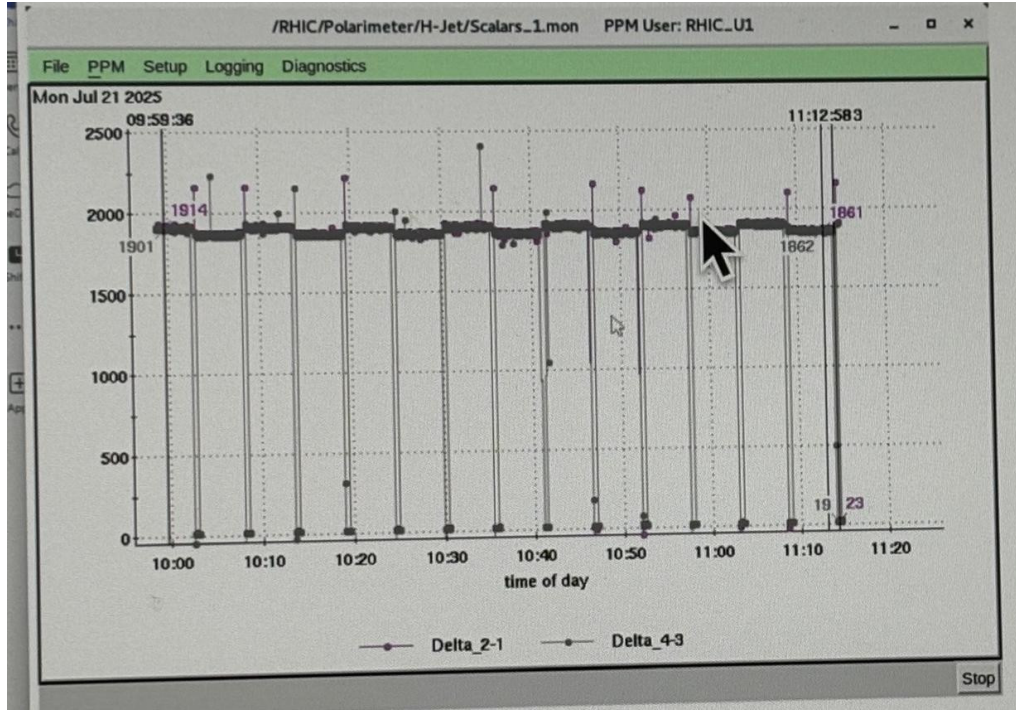


FIG. 8: HJET analysis code output. Apparently, the three signals are ABS WFT on – ABS SFT off, ABS WFT off – ABS SFT on, and both off.

a. Both ABS transitions off

$$\begin{pmatrix} n \\ n \\ n \\ n \end{pmatrix} \xrightarrow{\text{transmission A}} \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \quad (45)$$

b. Only ABS-SFT on

$$\begin{pmatrix} n \\ n \\ n \\ n \end{pmatrix} \xrightarrow{\text{transmission A}} \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{transition } f_{24}^A} \begin{pmatrix} n \sigma_1^A \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \\ n \sigma_3^A \\ n (\sigma_4^A \varepsilon_{24}^A + \sigma_2^A (1 - \varepsilon_{24}^A)) \end{pmatrix} \quad (46)$$

c. Only ABS-WFT on

$$\begin{pmatrix} n \\ n \\ n \\ n \end{pmatrix} \xrightarrow{\text{transmission A}} \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{transition } f_{13}^A} \begin{pmatrix} n (\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \\ n \sigma_2^A \\ n (\sigma_3^A \varepsilon_{13}^A + \sigma_1^A (1 - \varepsilon_{13}^A)) \\ n \sigma_4^A \end{pmatrix} \quad (47)$$

d. Both ABS transitions on

$$\begin{pmatrix} n \\ n \\ n \\ n \end{pmatrix} \xrightarrow{\text{transmission A}} \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{transitions } f_{13}^A, f_{24}^A} \begin{pmatrix} n (\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \\ n (\sigma_3^A \varepsilon_{13}^A + \sigma_1^A (1 - \varepsilon_{13}^A)) \\ n (\sigma_4^A \varepsilon_{24}^A + \sigma_2^A (1 - \varepsilon_{24}^A)) \end{pmatrix} \quad (48)$$

B. Transmission further from the target through the BRP

Each of these four cases further splits into four subcases as the atoms travel through the Breit-Rabi polarimeter.

1. ABS-WFT off, ABS-SFT off

a. BRP-WFT off, BRP-SFT off, signal I_0

$$\begin{aligned}
 & \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{transmission B1}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \\ n \sigma_2^A \sigma_2^{B1} \\ n \sigma_3^A \sigma_3^{B1} \\ n \sigma_4^A \sigma_4^{B1} \end{pmatrix} \xrightarrow{\text{blocker}} \\
 & \xrightarrow{\text{blocker}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \\ n \sigma_2^A \sigma_2^{B1} \\ 0 \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, transmission B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \sigma_1^{B2} \\ n \sigma_2^A \sigma_2^{B1} \sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix}
 \end{aligned} \tag{49}$$

b. BRP-WFT off, BRP-SFT on, signal I_1

$$\begin{aligned}
 & \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{transmission B1, blocker}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \\ n \sigma_2^A \sigma_2^{B1} \\ 0 \\ 0 \end{pmatrix} \xrightarrow{\text{transmission } f_{24}^B} \\
 & \xrightarrow{\text{transmission } f_{24}^B} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \\ n \sigma_2^A \sigma_2^{B1} \varepsilon_{24}^B \\ 0 \\ n \sigma_2^A \sigma_2^{B1} (1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \sigma_1^{B2} \\ n \sigma_2^A \sigma_2^{B1} \sigma_2^{B2} \varepsilon_{24}^B \\ 0 \\ 0 \end{pmatrix}
 \end{aligned} \tag{50}$$

c. BRP-WFT on, BRP-SFT off, signal I_2

$$\begin{aligned}
 & \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{13}^B} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \varepsilon_{13}^B \\ n \sigma_2^A \sigma_2^{B1} \\ n \sigma_1^A \sigma_1^{B1} (1 - \varepsilon_{13}^B) \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \sigma_1^{B2} \varepsilon_{13}^B \\ n \sigma_2^A \sigma_2^{B1} \sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix}
 \end{aligned} \tag{51}$$

d. BRP-WFT on, BRP-SFT on, signal I_3

$$\begin{aligned}
 & \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{24}^B, f_{13}^B} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \varepsilon_{13}^B \\ n \sigma_2^A \sigma_2^{B1} \varepsilon_{24}^B \\ n \sigma_1^A \sigma_1^{B1} (1 - \varepsilon_{13}^B) \\ n \sigma_2^A \sigma_2^{B1} (1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \sigma_1^{B2} \varepsilon_{13}^B \\ n \sigma_2^A \sigma_2^{B1} \sigma_2^{B2} \varepsilon_{24}^B \\ 0 \\ 0 \end{pmatrix}
 \end{aligned} \tag{52}$$

2. ABS-WFT off, ABS-SFT on

a. BRP-WFT off, BRP-SFT off, signal I_4

$$\begin{aligned}
& \begin{pmatrix} n \sigma_1^A \\ n \sigma_2^A \\ n \sigma_3^A \\ n \sigma_4^A \end{pmatrix} \xrightarrow{\text{transition } f_{24}^A} \begin{pmatrix} n \sigma_1^A \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \\ n \sigma_3^A \\ n (\sigma_4^A \varepsilon_{24}^A + \sigma_2^A (1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{transmission B1}} \\
& \xrightarrow{\text{transmission B1}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} \\ n \sigma_3^A \sigma_3^{B1} \\ n (\sigma_4^A \varepsilon_{24}^A + \sigma_2^A (1 - \varepsilon_{24}^A)) \sigma_4^{B1} \end{pmatrix} \xrightarrow{\text{blocker}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} \\ 0 \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \sigma_1^{B2} \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} \sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{53}$$

b. BRP-WFT off, BRP-SFT on, signal I_5

$$\begin{aligned}
& \begin{pmatrix} n \sigma_1^A \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \\ n \sigma_3^A \\ n (\sigma_4^A \varepsilon_{24}^A + \sigma_2^A (1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{24}^B} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} \varepsilon_{24}^B \\ 0 \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} (1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \sigma_1^{B1} \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \varepsilon_{24}^B \sigma_2^{B1} \sigma_2^{B1} \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{54}$$

c. BRP-WFT on, BRP-SFT off, signal I_6

$$\begin{aligned}
& \begin{pmatrix} n \sigma_1^A \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \\ n \sigma_3^A \\ n (\sigma_4^A \varepsilon_{24}^A + \sigma_2^A (1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{blocker, B1, } f_{13}^B} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \varepsilon_{13}^B \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} \\ n \sigma_1^A \sigma_1^{B1} (1 - \varepsilon_{13}^B) \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \varepsilon_{13}^B \sigma_1^{B2} \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} \sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{55}$$

d. BRP-WFT on, BRP-SFT on, signal I_7

$$\begin{aligned}
& \begin{pmatrix} n \sigma_1^A \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \\ n \sigma_3^A \\ n (\sigma_4^A \varepsilon_{24}^A + \sigma_2^A (1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{24}^B, f_{13}^B} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \varepsilon_{13}^B \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} \varepsilon_{24}^B \\ n \sigma_1^A \sigma_1^{B1} (1 - \varepsilon_{13}^B) \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^{B1} (1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n \sigma_1^A \sigma_1^{B1} \varepsilon_{13}^B \sigma_1^{B2} \\ n (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \varepsilon_{24}^B \sigma_2^{B1} \sigma_2^{B1} \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{56}$$

3. ABS-WFT on, ABS-SFT off

a. BRP-WFT off, WFT-SFT off, signal I_8

$$\begin{aligned}
& \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n\sigma_2^A \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n\sigma_4^A \end{pmatrix} \xrightarrow{\text{transmission B1}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1} \\ n\sigma_2^A\sigma_2^{B1} \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A))\sigma_3^{B1} \\ n\sigma_4^A\sigma_4^{B1} \end{pmatrix} \xrightarrow{\text{blocker}} \\
& \xrightarrow{\text{blocker}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1} \\ n\sigma_2^A\sigma_2^{B1} \\ 0 \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\sigma_1^{B2} \\ n\sigma_2^A\sigma_2^{B1}\sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{57}$$

b. BRP-WFT off, WFT-SFT on, signal I_9

$$\begin{aligned}
& \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n\sigma_2^A \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n\sigma_4^A \end{pmatrix} \xrightarrow{\text{blocker, B1, } f_{24}^B} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1} \\ n\sigma_2^A\sigma_2^{B1}\varepsilon_{24}^B \\ 0 \\ n\sigma_2^A\sigma_2^{B1}(1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\sigma_1^{B2} \\ n\sigma_2^A\sigma_2^{B1}\sigma_2^{B2}\varepsilon_{24}^B \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{58}$$

c. BRP-WFT on, WFT-SFT off, signal I_{10}

$$\begin{aligned}
& \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n\sigma_2^A \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n\sigma_4^A \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{13}^B} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\varepsilon_{13}^B \\ n\sigma_2^A\sigma_2^{B1} \\ n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}(1 - \varepsilon_{13}^B) \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\sigma_1^{B2}\varepsilon_{13}^B \\ n\sigma_2^A\sigma_2^{B1}\sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{59}$$

d. BRP-WFT on, WFT-SFT on, signal I_{11}

$$\begin{aligned}
& \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n\sigma_2^A \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n\sigma_4^A \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{24}^B, f_{13}^B} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\varepsilon_{13}^B \\ n\sigma_2^A\sigma_2^{B1}\varepsilon_{24}^B \\ n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}(1 - \varepsilon_{13}^B) \\ n\sigma_2^A\sigma_2^{B1}(1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
& \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\sigma_1^{B2}\varepsilon_{13}^B \\ n\sigma_2^A\sigma_2^{B1}\sigma_2^{B2}\varepsilon_{24}^B \\ 0 \\ 0 \end{pmatrix}
\end{aligned} \tag{60}$$

C. ABS-WFT on, ABS-SFT on

a. ABS-WFT on, ABS-SFT on, signal I_{12}

$$\begin{aligned}
 & \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A)) \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_4^A \varepsilon_{24}^A + \sigma_2^A(1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{transmission B1}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1} \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1} \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A))\sigma_3^{B1} \\ n(\sigma_4^A \varepsilon_{24}^A + \sigma_2^A(1 - \varepsilon_{24}^A))\sigma_4^{B1} \end{pmatrix} \xrightarrow{\text{blocker}} \\
 & \xrightarrow{\text{blocker}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1} \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1} \\ 0 \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
 & \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\sigma_1^{B2} \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}\sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix} \quad (61)
 \end{aligned}$$

b. BRP-WFT off, BRP-SFT on, signal I_{13}

$$\begin{aligned}
 & \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A)) \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_4^A \varepsilon_{24}^A + \sigma_2^A(1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{24}^B} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1} \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}\varepsilon_{24}^B \\ 0 \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}(1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
 & \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\sigma_1^{B2} \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}\sigma_2^{B2}\varepsilon_{24}^B \\ 0 \\ 0 \end{pmatrix} \quad (62)
 \end{aligned}$$

c. BRP-WFT on, BRP-SFT off, signal I_{14}

$$\begin{aligned}
 & \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A)) \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_4^A \varepsilon_{24}^A + \sigma_2^A(1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{13}^B} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\varepsilon_{13}^B \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1} \\ n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}(1 - \varepsilon_{13}^B) \\ 0 \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
 & \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\varepsilon_{13}^B\sigma_1^{B1}\sigma_1^{B2} \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}\sigma_2^{B2} \\ 0 \\ 0 \end{pmatrix} \quad (63)
 \end{aligned}$$

d. BRP-WFT on, BRP-SFT on signal I_{15}

$$\begin{aligned}
 & \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A)) \\ n(\sigma_3^A \varepsilon_{13}^A + \sigma_1^A(1 - \varepsilon_{13}^A)) \\ n(\sigma_4^A \varepsilon_{24}^A + \sigma_2^A(1 - \varepsilon_{24}^A)) \end{pmatrix} \xrightarrow{\text{B1, blocker, } f_{24}^B} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\varepsilon_{13}^B \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}\varepsilon_{24}^B \\ n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}(1 - \varepsilon_{13}^B) \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}(1 - \varepsilon_{24}^B) \end{pmatrix} \xrightarrow{\text{blocker, B2}} \\
 & \xrightarrow{\text{blocker, B2}} \begin{pmatrix} n(\sigma_1^A \varepsilon_{13}^A + \sigma_3^A(1 - \varepsilon_{13}^A))\sigma_1^{B1}\sigma_1^{B2}\varepsilon_{13}^B \\ n(\sigma_2^A \varepsilon_{24}^A + \sigma_4^A(1 - \varepsilon_{24}^A))\sigma_2^{B1}\sigma_2^{B2}\varepsilon_{24}^B \\ 0 \\ 0 \end{pmatrix} \quad (64)
 \end{aligned}$$

D. Signals in BRP

Since, in all cases, the blockers ensure that σ_k^{B1} and σ_k^{B2} always appear as a product, we can denote $\sigma_k^B = \sigma_k^{B1}\sigma_k^{B2}$.

TABLE IV: The BRP signals I_k written using all the transmissions, including σ_3^A and σ_4^A . Here, σ_k^A denote the transmissions through the ABS sextupole system, while σ_k^B denote those through the two BRP sextupoles.

Symbol	ABS-WFT	ABS-SFT	BRP-WFT	BRP-WFT	BRP signal
I_0	off	off	off	off	$s_1 = (\sigma_1^A \sigma_1^B + \sigma_2^A \sigma_2^B) n$
I_4	off	on	off	off	$s_5 = (\sigma_1^A \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B) n$
I_8	on	off	off	off	$s_9 = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B + \sigma_2^A \sigma_2^B) n$
I_{12}	on	on	off	off	$s_{13} = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B) n$
I_{14}	on	on	on	off	$s_{15} = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \varepsilon_{13}^B \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B) n$
I_1	off	off	off	on	$s_2 = (\sigma_1^A \sigma_1^B + \sigma_2^A \sigma_2^B \varepsilon_{24}^B) n$
I_2	off	off	on	off	$s_3 = (\sigma_1^A \sigma_1^B \varepsilon_{13}^B + \sigma_2^A \sigma_2^B) n$
I_3	off	off	on	on	$s_4 = (\sigma_1^A \sigma_1^B \varepsilon_{13}^B + \sigma_2^A \sigma_2^B \varepsilon_{24}^B) n$
I_5	off	on	off	on	$s_6 = (\sigma_1^A \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B \varepsilon_{24}^B) n$
I_6	off	on	on	off	$s_7 = (\sigma_1^A \sigma_1^B \varepsilon_{13}^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B) n$
I_7	off	on	on	on	$s_8 = (\sigma_1^A \sigma_1^B \varepsilon_{13}^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B \varepsilon_{24}^B) n$
I_9	on	off	off	on	$s_{10} = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B + \sigma_2^A \sigma_2^B \varepsilon_{24}^B) n$
I_{10}	on	off	on	off	$s_{11} = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B \varepsilon_{13}^B + \sigma_2^A \sigma_2^B) n$
I_{11}	on	off	on	on	$s_{12} = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B \varepsilon_{13}^B + \sigma_2^A \sigma_2^B \varepsilon_{24}^B) n$
I_{13}	on	on	off	on	$s_{14} = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B \varepsilon_{24}^B) n$
I_{15}	on	on	on	on	$s_{16} = ((\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B \varepsilon_{13}^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B \varepsilon_{24}^B) n$

E. Measurement strategy using system in Table IV

We treat the magnetic transmissions $\sigma_i^{A,B}$ as fixed constants, while the transition efficiencies $\varepsilon_{13}^A, \varepsilon_{24}^A, \varepsilon_{13}^B$ are under experimental control. Defining normalized signals $i_k = I_k/n$, the five measured configurations read

$$i_0 = \sigma_1^A \sigma_1^B + \sigma_2^A \sigma_2^B, \quad (65)$$

$$i_4 = \sigma_1^A \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B, \quad (66)$$

$$i_8 = (\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B + \sigma_2^A \sigma_2^B, \quad (67)$$

$$i_{12} = (\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B, \quad (68)$$

$$i_{14} = (\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \varepsilon_{13}^B \sigma_1^B + (\sigma_2^A \varepsilon_{24}^A + \sigma_4^A (1 - \varepsilon_{24}^A)) \sigma_2^B. \quad (69)$$

Among $\{i_0, i_4, i_8, i_{12}\}$ there is one linear dependence,

$$i_{12} = i_4 + i_8 - i_0, \quad (70)$$

so they provide only three independent equations. Turning on BRP–WFT yields

$$i_{14} - i_{12} = (\varepsilon_{13}^B - 1) (\sigma_1^A \varepsilon_{13}^A + \sigma_3^A (1 - \varepsilon_{13}^A)) \sigma_1^B, \quad (71)$$

which is an additional independent constraint whenever $\varepsilon_{13}^B \neq 1$ and the prefactor is nonzero.

F. Linearization in the controllable efficiencies and extraction strategy

Introduce four constant (unknown) combinations of transmissions:

$$B_1 = \sigma_3^A \sigma_1^B, \quad C_1 = (\sigma_1^A - \sigma_3^A) \sigma_1^B, \quad B_2 = \sigma_4^A \sigma_2^B, \quad C_2 = (\sigma_2^A - \sigma_4^A) \sigma_2^B. \quad (72)$$

With these, the signals become linear in the controllable ε 's:

$$i_8(\varepsilon_{13}^A) = \underbrace{(B_1 + \sigma_2^A \sigma_2^B)}_{\text{const.}} + C_1 \varepsilon_{13}^A, \quad (73)$$

$$i_4(\varepsilon_{24}^A) = \underbrace{(B_1 + C_1)}_{\text{const.}} + (B_2 + C_2 \varepsilon_{24}^A), \quad (74)$$

$$i_{12}(\varepsilon_{13}^A, \varepsilon_{24}^A) = (B_1 + C_1 \varepsilon_{13}^A) + (B_2 + C_2 \varepsilon_{24}^A), \quad (75)$$

$$i_{14}(\varepsilon_{13}^A, \varepsilon_{24}^A, \varepsilon_{13}^B) = \varepsilon_{13}^B (B_1 + C_1 \varepsilon_{13}^A) + (B_2 + C_2 \varepsilon_{24}^A). \quad (76)$$

a. How it works experimentally.

1. **Determine C_1 (from the ABS–WFT transition).** Keep ε_{24}^A fixed and vary the efficiency of the ABS weak–field transition ε_{13}^A (e.g. between 0 and 1). From the linearized form

$$i_8(\varepsilon_{13}^A) = (B_1 + \sigma_2^A \sigma_2^B) + C_1 \varepsilon_{13}^A, \quad (77)$$

the slope in ε_{13}^A gives C_1 . Equivalently, one may use

$$i_{12}(\varepsilon_{13}^A) = (B_1 + C_1 \varepsilon_{13}^A) + (B_2 + C_2 \varepsilon_{24}^A), \quad (78)$$

or

$$i_{14}(\varepsilon_{13}^A) = \varepsilon_{13}^B (B_1 + C_1 \varepsilon_{13}^A) + (B_2 + C_2 \varepsilon_{24}^A), \quad (79)$$

where in the last case the slope is $\varepsilon_{13}^B C_1$; division by the known ε_{13}^B yields C_1 .

2. **Determine C_2 (from the ABS–SFT transition).** Keep ε_{13}^A fixed and vary the efficiency of the ABS strong–

field transition ε_{24}^A . From

$$i_4(\varepsilon_{24}^A) = (B_1 + C_1) + (B_2 + C_2 \varepsilon_{24}^A), \quad (80)$$

the slope in ε_{24}^A is C_2 . The same slope can also be extracted from i_{12} or i_{14} as a function of ε_{24}^A .

3. Determine B_1 and B_2 . Once C_1 and C_2 are known, the intercepts provide B_1 and B_2 . For example:

$$i_8(\varepsilon_{13}^A = 0) = B_1 + \sigma_2^A \sigma_2^B, \quad (81)$$

$$i_4(\varepsilon_{24}^A = 0) = (B_1 + C_1) + B_2. \quad (82)$$

Together with the constraint

$$i_0 = (B_1 + C_1) + (B_2 + C_2), \quad (83)$$

these relations allow the unique determination of B_1 and B_2 .

4. Role of BRP–WFT. With the BRP weak-field transition active,

$$i_{14} - i_{12} = (\varepsilon_{13}^B - 1)(B_1 + C_1 \varepsilon_{13}^A), \quad (84)$$

which isolates the part of the signal governed by the ABS weak-field transition. This provides an additional independent constraint whenever $\varepsilon_{13}^B \neq 1$, improving the extraction of B_1 and C_1 .

1. Closed-form extraction of the transition efficiencies

From the linearized system and the definitions of C_1 and C_2 , the transition efficiencies follow directly as

$$\varepsilon_{13}^A = 1 + \frac{i_8 - i_0}{C_1}, \quad (85)$$

$$\varepsilon_{24}^A = 1 + \frac{i_4 - i_0}{C_2}. \quad (86)$$

Equivalently, without using i_0 explicitly,

$$\varepsilon_{13}^A = \frac{i_8 - (B_2 + C_2) - B_1}{C_1}, \quad \varepsilon_{24}^A = \frac{i_4 - (B_1 + C_1) - B_2}{C_2}. \quad (87)$$

When BRP–WFT is active with a known $\varepsilon_{13}^B \neq 1$,

$$\varepsilon_{13}^A = \frac{\frac{i_{14} - i_{12}}{\varepsilon_{13}^B - 1} - B_1}{C_1}, \quad \varepsilon_{24}^A = \frac{i_{12} - (B_1 + C_1 \varepsilon_{13}^A) - B_2}{C_2}. \quad (88)$$

a. Outcome. This linear strategy exploits the fact that σ 's are fixed and only ε 's are tunable: by scanning ε_{13}^A and ε_{24}^A you obtain straight-line fits whose slopes yield C_1 and C_2 , and whose intercepts (together with i_0) determine B_1 and B_2 . Once B_1, B_2, C_1, C_2 are known, the *true* efficiencies for any subsequent settings follow directly from the linear forms above, with standard (Gaussian) error propagation.

G. Recovery of individual transmissions using the absolute transmissions σ_1^A and σ_2^A

a. Absolute inputs. The absolute transmissions σ_1^A and σ_2^A of the two atomic branches, obtained for example from the Monte-Carlo simulations, are treated as fixed inputs.

b. Result. With the absolute inputs σ_1^A and σ_2^A known, the remaining transmissions follow uniquely from the fitted constants B_1, C_1, B_2, C_2 :

$$\sigma_1^B = \frac{B_1 + C_1}{\sigma_1^A}, \quad \sigma_3^A = \frac{B_1 \sigma_1^A}{B_1 + C_1}, \quad (89)$$

$$\sigma_2^B = \frac{B_2 + C_2}{\sigma_2^A}, \quad \sigma_4^A = \frac{B_2 \sigma_2^A}{B_2 + C_2}. \quad (90)$$

c. Consistency. The identity

$$i_0 = \sigma_1^A \sigma_1^B + \sigma_2^A \sigma_2^B = (B_1 + C_1) + (B_2 + C_2) \quad (91)$$

provides an internal check of the solution.

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Appendix A: Traffic flow covariance problem

1. Setup

Consider a road segment where we measure traffic at two points: the entrance and the exit. Let A denote the number of cars entering the road (a random variable), and let B denote the number of cars exiting at the end point (also a random variable). Between these two measurement points, there is an exit ramp where cars may leave, but no additional cars can enter the road.

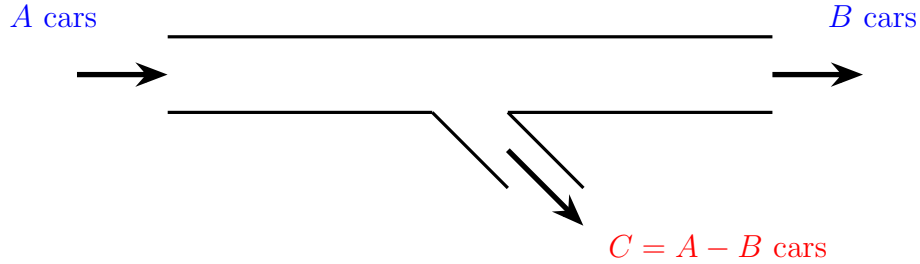


FIG. 9: Road segment with entrance (A cars), exit ramp (C cars), and exit point (B cars).

Each car that enters has a constant probability p of exiting via the intermediate exit ramp, and thus a probability $1 - p$ of continuing all the way to the exit point where B is measured. We are interested in understanding the relationship between A and B through their covariance.

2. The conditional distribution

We introduce the notation $B | A$, which reads as “ B given A ” or “ B conditional on A .” Even when we know exactly how many cars entered (the value of A), the number reaching the exit point (B) remains random because each car independently makes the decision to exit or continue.

For instance, if $A = 10$ cars enter and $p = 0.3$, then $B \mid A = 10$ can take any value from 0 to 10. Each of the 10 cars reaches the exit point with probability 0.7, so we expect around 7 cars to make it through, but the actual number varies randomly. We might observe $B = 5$, or $B = 8$, or any other value between 0 and 10.

Given A cars enter, the number reaching the exit follows a binomial distribution

$$B \mid A \sim \text{Binomial}(A, 1 - p). \quad (\text{A1})$$

This means that each of the A cars independently reaches the exit point with probability $1 - p$. The binomial distribution $\text{Binomial}(n, q)$ has well-known properties: its expected value is nq and its variance is $nq(1 - q)$. Applying these formulas to our case with $n = A$ trials and success probability $q = 1 - p$, we obtain

$$E[B \mid A] = A(1 - p), \quad \text{Var}(B \mid A) = A(1 - p)p. \quad (\text{A2})$$

The expected value tells us that if A cars enter and each has probability $1 - p$ of reaching the exit, we expect on average $A(1 - p)$ cars to make it through. The variance measures the spread around this expected value, which depends on both the number of cars and the probabilities involved. It is important to note that $E[B \mid A]$ is itself a random variable because it depends on the random variable A .

3. The Binomial distribution

For a random variable $X \sim \text{Binomial}(n, q)$, the probability mass function gives the probability of observing exactly k successes in n independent trials, where each trial succeeds with probability q

$$P(X = k) = \binom{n}{k} q^k (1 - q)^{n-k} \quad \text{for } k = 0, 1, 2, \dots, n. \quad (\text{A3})$$

Here, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient representing the number of ways to choose k successes from n trials. The term q^k gives the probability of k successes, while $(1 - q)^{n-k}$ gives the probability of the remaining $n - k$ failures.

In our traffic context, this becomes

$$P(B = k \mid A) = \binom{A}{k} (1 - p)^k p^{A-k} \quad \text{for } k = 0, 1, 2, \dots, A. \quad (\text{A4})$$

To make this concrete, consider again $A = 10$ and $p = 0.3$. Then $B \mid A = 10 \sim \text{Binomial}(10, 0.7)$, and

$$P(B = k \mid A = 10) = \binom{10}{k} (0.7)^k (0.3)^{10-k}. \quad (\text{A5})$$

Computing a few specific probabilities illustrates the distribution

$$P(B = 0) = \binom{10}{0} (0.7)^0 (0.3)^{10} \approx 0.000006 \quad (\text{A6})$$

$$P(B = 5) = \binom{10}{5} (0.7)^5 (0.3)^5 \approx 0.103 \quad (\text{A7})$$

$$P(B = 7) = \binom{10}{7} (0.7)^7 (0.3)^3 \approx 0.267 \quad (\text{most likely}) \quad (\text{A8})$$

$$P(B = 10) = \binom{10}{10} (0.7)^{10} (0.3)^0 \approx 0.028 \quad (\text{A9})$$

The distribution is centered around the expected value $E[B] = 10 \times 0.7 = 7$, with the probability mass concentrated near this value.

4. Computing the covariance

To find the covariance between A and B , we employ the law of total covariance, which decomposes the covariance into two components,

$$\text{cov}(A, B) = E[\text{cov}(A, B \mid A)] + \text{cov}(E[B \mid A], A). \quad (\text{A10})$$

The first term, $E[\text{cov}(A, B \mid A)]$, captures the covariance between A and B within each fixed value of A . However, when we condition on A , the variable A becomes a constant, and the covariance between a random variable and a constant is always zero. Therefore

$$E[\text{cov}(A, B \mid A)] = E[0] = 0. \quad (\text{A11})$$

The second term, $\text{cov}(E[B \mid A], A)$, captures how the expected value of B varies with A . Since $E[B \mid A] = A(1 - p)$, we have

$$\text{cov}(E[B \mid A], A) = \text{cov}(A(1 - p), A) = (1 - p) \cdot \text{cov}(A, A). \quad (\text{A12})$$

A crucial identity in probability theory is that the covariance of a variable with itself equals its variance: $\text{cov}(A, A) = \text{Var}(A)$. This follows directly from the definition of covariance. Substituting this identity yields

$$\text{cov}(A, B) = (1 - p) \cdot \text{Var}(A). \quad (\text{A13})$$

We have established that the covariance between the number of cars entering and the number exiting is

$$\text{cov}(A, B) = (1 - p) \cdot \text{Var}(A) \quad (\text{A14})$$

This result is intuitive. The covariance is positive, reflecting the fact that more cars entering generally leads to more cars exiting. The magnitude is proportional to the variance of the entrance count, scaled by the pass-through rate $(1 - p)$. When p is small (few cars exit via the intermediate ramp), the covariance approaches $\text{Var}(A)$, indicating a strong relationship. When p is large (most cars exit before reaching the measurement point), the covariance becomes smaller, reflecting the weaker relationship between entrance and exit counts.

5. Transmission and error propagation

In traffic engineering, the transmission of a road segment is defined as the ratio of cars exiting to cars entering

$$T = \frac{B}{A}. \quad (\text{A15})$$

This quantity measures the fraction of entering traffic that successfully reaches the exit point. In our model, we expect the transmission to be approximately $1 - p$, since each car independently has probability $1 - p$ of reaching the exit. However, because both B and A are random variables, the transmission T is itself random and exhibits variability.

a. Expected value of the transmission

To find $E[T] = E[B/A]$, we use the law of total expectation

$$E[T] = E\left[\frac{B}{A}\right] = E\left[E\left[\frac{B}{A} \mid A\right]\right]. \quad (\text{A16})$$

Given A , we know that $E[B \mid A] = A(1 - p)$, so

$$E\left[\frac{B}{A} \mid A\right] = \frac{E[B \mid A]}{A} = \frac{A(1 - p)}{A} = 1 - p. \quad (\text{A17})$$

Therefore

$$E[T] = E[1 - p] = 1 - p. \quad (\text{A18})$$

The expected transmission equals the pass-through probability, as intuition suggests.

b. Variance of the transmission

Computing the variance of a ratio is more complex. We use the law of total variance

$$\text{Var}(T) = E[\text{Var}(T | A)] + \text{Var}(E[T | A]). \quad (\text{A19})$$

For the first term, given A , we have $T | A = B/A$ where $B | A \sim \text{Binomial}(A, 1 - p)$. The variance of $B | A$ is $A(1 - p)p$, so

$$\text{Var}(T | A) = \text{Var}\left(\frac{B}{A} \middle| A\right) = \frac{1}{A^2} \text{Var}(B | A) = \frac{A(1 - p)p}{A^2} = \frac{(1 - p)p}{A}. \quad (\text{A20})$$

Taking the expectation

$$E[\text{Var}(T | A)] = E\left[\frac{(1 - p)p}{A}\right] = (1 - p)p \cdot E\left[\frac{1}{A}\right]. \quad (\text{A21})$$

For the second term, since $E[T | A] = 1 - p$ is constant (does not depend on A), we have

$$\text{Var}(E[T | A]) = \text{Var}(1 - p) = 0. \quad (\text{A22})$$

Therefore, the variance of the transmission is

$$\text{Var}(T) = (1 - p)p \cdot E\left[\frac{1}{A}\right] \quad (\text{A23})$$

c. Delta method approximation

The variance of the transmission depends on $E[1/A]$, the expected value of the reciprocal of the number of entering cars. This quantity is difficult to compute exactly without knowing the distribution of A , but we can approximate it using the delta method. The delta method uses a Taylor expansion of the function $g(A) = 1/A$ around the mean $\mu_A = E[A]$. Taking the Taylor expansion to second order

$$g(A) = g(\mu_A) + g'(\mu_A)(A - \mu_A) + \frac{1}{2}g''(\mu_A)(A - \mu_A)^2 + \dots \quad (\text{A24})$$

For $g(A) = 1/A$, we compute the derivatives

$$g(\mu_A) = \frac{1}{\mu_A} \quad (\text{A25})$$

$$g'(A) = -\frac{1}{A^2}, \quad g'(\mu_A) = -\frac{1}{\mu_A^2} \quad (\text{A26})$$

$$g''(A) = \frac{2}{A^3}, \quad g''(\mu_A) = \frac{2}{\mu_A^3}. \quad (\text{A27})$$

Substituting into the Taylor expansion

$$\frac{1}{A} \approx \frac{1}{\mu_A} - \frac{1}{\mu_A^2}(A - \mu_A) + \frac{1}{2} \cdot \frac{2}{\mu_A^3}(A - \mu_A)^2. \quad (\text{A28})$$

Taking expectations and using $E[A - \mu_A] = 0$ and $E[(A - \mu_A)^2] = \sigma_A^2$, we obtain

$$E\left[\frac{1}{A}\right] \approx \frac{1}{\mu_A} + \frac{\sigma_A^2}{\mu_A^3}. \quad (\text{A29})$$

This is a second-order approximation. If the coefficient of variation σ_A/μ_A is small, we can use the simpler first-order approximation $E[1/A] \approx 1/\mu_A$, yielding

$$\text{Var}(T) \approx \frac{(1-p)p}{\mu_A}. \quad (\text{A30})$$

This shows that the variability in transmission decreases as the average number of entering cars increases. With more cars, the law of large numbers ensures that the observed transmission ratio B/A concentrates more tightly around its expected value $1-p$. The variance also depends on $p(1-p)$, which is maximized when $p = 0.5$ and decreases as p approaches 0 or 1.

d. Standard error of the transmission

The standard error of the transmission, which measures the typical deviation from the expected transmission, is approximately

$$\sqrt{\text{Var}(T)} \approx \sqrt{\frac{(1-p)p}{\mu_A}}. \quad (\text{A31})$$

This quantity is useful for constructing confidence intervals or assessing the precision of transmission measurements in practice.

e. Estimating transmission variance from data

In practice, we often do not know the exit probability p in advance. Instead, we observe a single realization where at the entrance we measure A cars entering and simultaneously B cars exiting. From this single pair of observations, we must estimate both the transmission and assess its uncertainty. The observed transmission is simply

$$\hat{T} = \frac{B}{A}. \quad (\text{A32})$$

From this, we can estimate the exit probability as

$$\hat{p} = 1 - \hat{T} = 1 - \frac{B}{A} = \frac{A - B}{A}. \quad (\text{A33})$$

To assess the uncertainty in this transmission estimate, we substitute $\hat{p}$ into our variance approximation. However, we face a fundamental challenge: the variance formula

$$\text{Var}(T) \approx \frac{(1-p)p}{\mu_A} \quad (\text{A34})$$

requires knowledge of $\mu_A = E[A]$, the expected number of entering cars. With only a single observation of A , we must either assume a value for μ_A based on prior knowledge (such as typical traffic volumes), or treat the observed value A as a proxy for μ_A if we believe traffic conditions are stable.

Using the observed values as proxies, the estimated variance becomes

$$\widehat{\text{Var}}(T) \approx \frac{(1-\hat{p})\hat{p}}{A} = \frac{\hat{T}(1-\hat{T})}{A} = \frac{\frac{B}{A}(1-\frac{B}{A})}{A} = \frac{B(A-B)}{A^3}. \quad (\text{A35})$$

The estimated standard error is then

$$\widehat{\text{SE}}(T) = \sqrt{\frac{B(A-B)}{A^3}} = \frac{\sqrt{B(A-B)}}{A^{3/2}} = \sqrt{\frac{\hat{T}(1-\hat{T})}{A}}. \quad (\text{A36})$$

This provides a measure of the uncertainty in the transmission estimate based on the single pair of observations. However, it is important to recognize the limitations: this estimate assumes that the observed traffic conditions are representative of typical conditions, and it does not account for potential variability in p over time or systematic measurement errors at the stations.

The standard error formula reveals several intuitive features. The uncertainty decreases with larger A (more entering cars provide better estimates), and it is maximized when $B \approx A/2$ (i.e., when $\hat{p} \approx 0.5$), corresponding to maximum uncertainty about which cars exit. When nearly all cars exit ($B \approx 0$) or nearly all continue ($B \approx A$), the uncertainty is smaller because the transmission is more extreme and easier to estimate precisely.